

Evaluating design-stage manufacturability decisions against the resolution of production

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Abstract

Design-stage manufacturability decisions increasingly rest on simulation and machine learning, yet whether a prediction is fit to decide is rarely evaluated against production. A framework is proposed whose unit, the production floor, is the difference that drift and scatter produce between two batches of one design within a run. A rule returning meets, fails or trial is characterised by a decisive distance in floors, beyond which at least 95 % of its verdicts are correct, and a safe distance, beyond which at most 5 % are wrong, for each relation of a new design to the produced evidence: a variant, a new process setting or a new family. A procedure turns them into a guard band for single verdicts. Thirteen rules were evaluated at a 95 % conformance level on open data of 18 deep-drawn design–process alternatives of 500 parts each, mostly process settings on one tool, with matched simulations; one floor of draw-in is 0.05 mm. With a produced sibling the nearest produced setting decided within 3.4 floors; for a new process setting no rule decided closer than 8.5 floors; in two transfers to a new family none was safe closer than 16. Within a family the choice of rule mattered more than the relation, and requirements at a performance index of 1.33 lay inside every decisive distance. With six to nine produced settings per fit, the simulation, as used, helped only across families, and learned models added abstention rather than accuracy. The framework gives evaluations of AI-based manufacturability support a production-referenced standard.

Keywords: design for manufacturing, machine learning, design decision making, verification and validation, operating characteristic, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming, as in most processes, the decision rests on a finite-element simulation ~~of the process~~ (Banabic, 2010): the ~~design team simulates the alternative;~~ team compares the predicted characteristics with the requirements ~~on the drawing and either releases the design~~ and releases the alternative, changes it or sends it to a physical trial. Simulation has moved such decisions forward ~~in the design process~~ (Thomke, 1998; Thomke and Fujimoto, 2000) (Thomke, 1998), and machine learning ~~has made~~ makes the loop faster ~~and broader~~ —surrogates—: surrogates replace simulations, ~~data-driven models learn from earlier products, and manufacturability checks are increasingly automated~~ (Yan et al, 2026; Liu et al, 2026) ~~—but neither models learn manufacturability from earlier designs, and checks are automated~~ (Yan et al, 2026; Ghadai et al, 2018; Zhang et al, 2018). Neither has changed the question ~~a designer~~ the team has to answer: *can this prediction be ~~trusted~~ relied on to decide?* A simulation that is right on average can ~~still~~ be wrong for the alternative at hand, and a model ~~that learned from~~ learned on earlier products may be asked about a design unlike any of them. Whitney argued in this journal that mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996), ~~and variation in production remains the reason~~ (Thornton et al, 2000): ~~the credibility of—~~ production variation, with which development organisations still contend (Thornton et al, 2000), adds a further reason. Whether a simulation-based decision is ~~ultimately credible is in the end~~ a property of production, and production evidence is ~~exactly~~ what is scarce at the design stage.

This paper treats the design-stage manufacturability decision itself as the object of study ~~and asks three questions~~. How can the fitness of a design-stage prediction for ~~a manufacturability~~ such a decision be expressed ~~independently of the predictor and of the~~ on a common production-referenced scale, for any predictor and characteristic (RQ1)? How does ~~that fitness~~ it depend on the ~~production evidence available on its amount~~ amount of production evidence and on how the new design relates to what has been produced (RQ2)? And what do simulation and machine learning contribute ~~to such a decision~~ once production evidence exists (RQ3)? ~~The questions sit between three strands of design research: decision-based design, which treats design as a sequence of decisions under uncertainty~~ (Hazelrigg, 1998; Lewis et al, 2006); ~~the margins that designers keep between predictions and requirements~~ (Eckert et al, 2019; Brahma and Wynn, 2020) ~~;~~ —Decision-based design (Hazelrigg, 1998; Lewis et al, 2006), design margins (Eckert et al, 2019; Brahma and Wynn, 2020) and the planning of verification and testing ~~in product development~~ (Thomke and Bell, 2001; Salado and Kannan, 2018; Tahera et al, 2019) ~~. Each needs all need~~ to know how far a prediction can be ~~trusted~~ relied on before a test is cheaper than a decision, ~~and none has a production-referenced measure of it.~~

~~A framework that provides such a measure is proposed. Its unit is~~ The framework proposed here has the *production floor* as its unit; the difference that drift and scatter produce between two batches of consecutive parts of one ~~and the same design that drift and scatter produce at a chosen confidence~~ design within a run, a floor in the

sense of a noise floor rather than of the shop floor. A rule that turns the design-stage evidence into ~~one of three verdicts—the a verdict (the~~ alternative meets the requirement, *fails* it, or goes to a physical *trial*—~~is~~) is characterised by two operating characteristics ~~expressed in floors: its decisive distance, beyond which at least 95 % of its verdicts are correct, in floors, its decisive distance~~ and its *safe distance*, beyond which at most 5 % are wrong. Both are functions of how far the requirement lies from the truth, in the way the operating characteristic of an acceptance plan is a function of the true quality, and between them the rule abstains rather than errs. They are evaluated separately for each *evidence relation* between a new design and the produced alternatives—a variant whose process setting has been produced with another, unresolvable variation; a new process setting within or outside the produced range; and a new design family—together with a calibration budget, an applicability guard and a six-step procedure alternatives, and a procedure turns them into a guard band for single verdicts. The decisions in scope are those of a team that has produced variants of a design family, in process planning and tryout and when production knowledge is carried to a new family; early product design is covered only by the new-family relation.

The ~~framework is evaluated on open data that make such an evaluation possible~~ evaluation uses open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design-process alternatives of 500 consecutive parts each, ~~scanned after drawing and after cutting (Baum and Heinzelmänn, 2026), together (Baum and Heinzelmänn, 2026), with finite-element simulations of the same process and tools (Baum and Heinzelmänn, 2025). Each produced (Baum and Heinzelmänn, 2025). Within each of two cup geometries the alternatives differ in blank-holder force and lubrication on one tool, so two of the three relations concern process settings, and the geometries give two transfers between families. Each alternative is judged as a new design—by leaving it out alone, or with every alternative that shares its process setting, or with its whole family—and design, and the verdicts of twelve rules thirteen rules at a 95 % conformance level~~ are compared with what its 500 parts *actually* did. In the terms of the design research methodology (Blessing and Chakrabarti, 2009), ~~the paper this~~ is a prescriptive study ~~whose framework and procedure are given an initial evaluation on production data; that evaluation establishes with an initial support evaluation: it shows~~ whether the framework discriminates between rules and evidence situations, not ~~how design teams use it whether design teams decide better with it, and its findings on one dataset are hypotheses for confirmation.~~

The contributions are fourfold. The first is the framework itself ~~the framework~~ (Section 3): ~~the production floor with its effect-to-scatter ratio and minimum resolvable change, and the decisive and safe distances as operating characteristics, with their assumptions, properties and an exact estimator. The second is:~~ an evaluation design for design-stage decision rules that separates the evidence relations by grouped cross-validation (Roberts et al, 2017) and isolates what the simulation ~~contributes by evaluating every calibrated rule with and without it adds~~ (Section 4.4). The third is the empirical answer to the three questions on deep drawing, including what the production floor contains and how the conclusions depend on the analysis choices

4); the empirical answers for deep drawing (Section 5). The fourth is a procedure and guidance for design practice that separate what is general from what holds for this dataset; and guidance and a reporting standard for evaluating simulation- and AI-based manufacturability support (Section 6).

Section 2 positions the work; Section 3 defines the framework; Section 4 describes the data, the characteristics and the evaluation design; Section 5 reports the results; Section 6 discusses what they establish and the threats to validity; and, Section 7 concludes, and Appendix A gives computation details and additional tables the notation, the robustness results and computation details.

2 Background and related work

2.1 Manufacturability evaluation and machine learning in design for manufacturing

Design for manufacturing (DFM) has long codified codifies what a process can make into rules, guidelines and cost models applied at the design stage (Boothroyd et al, 2010; Bralla, 1999)(Boothroyd et al, 2010), and quantitative manufacturability models have been used in this journal to drive design decisions (La Trobe-Bateman and Wild, 2003). Process simulation now predicts the outcome of a specific alternative rather than of a class of features, and machine Machine learning learns manufacturability from historical designs and their production records, from CAD models (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), from process data and simulated design spaces, or from both (Yan et al, 2026; Ballegeer and Benoit, 2026). Work in this journal has used data and models to choose between manufacturing processes from shape (Ben Slama et al, 2024), to quantify design robustness (Li et al, 2026), to reason about failure causes in design FMEA (Liu et al, 2026) and to show that measures left out of an alternative selection problem change which alternative is chosen (Eaton et al, 2026). A recurring difficulty is transfer: a model learned on one process, machine or product family is applied to another (Safdar et al, 2024), and its reproducibility and reusability are open questions (Xie et al, 2025, 2026). What these studies rarely have is a production record of the very alternatives the model is asked about, so their evaluation stops at prediction accuracy and does not reach the design decision; open datasets that would allow more are increasingly recognised as research contributions in their own right (Ahmed et al, 2025)(Yan et al, 2026), and generative models propose designs directly (Regenwetter et al, 2022); transfer between processes and product families remains difficult (Safdar et al, 2024). Learned manufacturability models are typically evaluated by held-out accuracy on labelled designs (Ghadai et al, 2018; Zhang et al, 2018; Williams et al, 2019), not by the decisions they support against the production of the alternatives they are asked about.

2.2 Design decisions under uncertainty, design types and their evaluation production evidence

Decision-based design casts design as choosing between alternatives whose outcomes are uncertain, with a preference over them (Hazelrigg, 1998; Lewis et al, 2006). In practice the uncertainty of a prediction a choice between alternatives with uncertain outcomes (Hazelrigg, 1998; Lewis et al, 2006). How much a team knows about a new alternative depends on the kind of design, variant, adaptive or original (Pahl et al, 2007), organised across products by product-family and platform design (Simpson et al, 2006; Jiao et al, 2007). The evidence relations of this paper are their counterparts for production evidence: a new variant of a produced setting, a new process setting and a new family. Production knowledge enters design through process-capability data (Thornton, 2004; Tata and Thornton, 1999) and tolerance analysis (Chase and Parkinson, 1991). Unresolved uncertainty is absorbed by margins : buffers between what is predicted and what is required, which are seldom made explicit and whose value is rarely known (Eckert et al, 2019; Brahma and Wynn, 2020). Testing is (Eckert et al, 2019; Brahma and Wynn, 2020), and the part of a margin that covers uncertainty is distinct from the part that absorbs change (Eckert and Isaksson, 2017). Tests and prototypes, the alternative to a margin, and its planning when to test, how many alternatives to test in parallel or in sequence, and which verification activities to perform has been are modelled as a trade between the cost costs of tests and the cost of errors (Thomke, 1998; Thomke and Bell, 2001; Loch et al, 2001; Salado and Kannan, 2018), with physical and virtual testing of errors (Thomke, 1998; Thomke and Bell, 2001; Salado and Kannan, 2018) and studied as design activities in their own right (Tahera et al, 2019). Variation risk management identifies the key characteristics whose variation matters and the processes that control them (Thornton, 1999; Thornton et al, 2000). The framework of this paper supplies a quantity these approaches assume but do not measure : how close to a requirement (Tahera et al, 2019; Camburn et al, 2017). When upstream information is fit to act on is the question of overlapped development (Krishnan et al, 1997; Terwiesch et al, 2002), and set-based concurrent engineering codifies production knowledge as feasibility limits for new designs (Sobek et al, 1999). All of these assume a measure of how far a design-stage prediction can decide, be relied on; the framework supplies one in a unit set by production, and therefore how large a margin has to be, or when a test is unavoidable. The evaluation of design methods themselves has its own methodology: the design research methodology separates the development of support from its evaluation (Blessing and Chakrabarti, 2009), the validation square distinguishes the structural and performance validity of a method (Seepersad et al, 2006), and evaluation designs from medicine have been proposed for design methods (Frey and Dym, 2006).

2.3 Simulation credibility, calibration and validation metrics

Verification and validation methodology asks ask whether a model is accurate enough for its intended use, and answers it by comparing predictions with experiments under

231 a stated validation metric (Oberkampf and Roy, 2010; Roy and Oberkampf, 2011)
 232 ; standards for computational solid mechanics and for the credibility of decisions
 233 based on models formalise the same question (ASME, 2020; NASA, 2024).
 234 Validation metrics compare predicted and observed distributions in physical
 235 units (Ferson et al, 2008; Liu et al, 2011), and predictive maturity tracks how the
 236 predictive capability of a calibrated model improves with validation experiments
 237 (Hemez et al, 2010) the distinction between its validation and application domains
 238 (Oberkampf and Roy, 2010) is the question that the evidence relation asks of
 239 a design; risk-informed credibility assessment scales the required evidence with
 240 the consequence of a wrong decision (ASME, 2018). Bayesian calibration adds
 241 a model of the discrepancy between simulator and reality, typically a Gaussian
 242 process over the inputs, together with calibration parameters of the simulator
 243 (Kennedy and O’Hagan, 2001; Higdon et al, 2008; Bayarri et al, 2007); discrepancy
 244 model and calibration parameters to a simulator (Kennedy and O’Hagan, 2001)
 245 ; multi-fidelity models scale a cheap model to an expensive one
 246 (Kennedy and O’Hagan, 2000), and omitting the discrepancy biases the calibrated
 247 parameters (Brynjarsdóttir and O’Hagan, 2014), and the identifiability of discrepancy
 248 and parameters limits what few experiments can teach (Arendt et al, 2012).
 249 Where a predictive parameters (Brynjarsdóttir and O’Hagan, 2014); where
 250 a model may be trusted relied on at all is the question of its valid input
 251 domain (Malak and Paredis, 2010). Conformal prediction gives distribution-
 252 free intervals with a finite-sample coverage guarantee under exchangeability
 253 (Vovk et al, 2005; Lei et al, 2018; Barber et al, 2021; Angelopoulos and Bates, 2023)
 254 ; including for quantiles (Romano et al, 2019) and under a shift of the inputs
 255 (Tibshirani et al, 2019) , and it has begun to appear in engineering design and quality
 256 assurance (Zong et al, 2026; Heddoub et al, 2026). These methods produce intervals
 257 and accuracy statements; how close to a requirement such an interval can be trusted
 258 to decide, and how much production evidence that takes, remains open.
 259 under exchangeability (Lei et al, 2018; Barber et al, 2021), also under
 260 covariate shift (Tibshirani et al, 2019) and for the control of decision
 261 risks (Angelopoulos et al, 2024). In sheet-metal forming, the accuracy of
 262 finite element predictions springback above all has been studied for decades
 263 (Wagoner et al, 2013; Chen et al, 2022), with friction (Hol et al, 2012) and the
 264 unloading behaviour of the material (Yoshida and Uemori, 2002) are among the
 265 main sources of error. Simulations are used to compensate springback in the tool
 266 design (Gan and Wagoner, 2004), robust process design under material and process
 267 scatter is well established (Strano, 2006; Hou et al, 2010; Wiebenga et al, 2014),
 268 often with metamodels (Jakumeit et al, 2005; Harsch et al, 2016), and data-driven
 269 models predict forming outcomes from process data in series production
 270 (Havinga et al, 2020; Purr et al, 2015). Simulation simulation error; simulation
 271 datasets of deep drawing have been released for support surrogate and transfer learn-
 272 ing (Baum and Heinzelmänn, 2025; Heinzelmänn et al, 2025a), with surrogates trained
 273 on them (Heinzelmänn et al, 2025b; Baum et al, 2025) and the gap between synthetic
 274 and real data identified as the obstacle to their industrial use (Baum et al, 2026b).
 275 The dataset’s authors decomposed and the deviation between simulation and parts
 276

and found that the part geometry explains 77–92 % of its variance and that gradient boosting with process measurements explains 81–92 % (Baum et al, 2026a); the present paper asks the complementary design question of how has been decomposed statistically (Baum et al, 2026a). How close to a requirement a decision taken before production, without process measurements of the new alternative, can be trusted such intervals and accuracy statements can be relied on to decide, and how much production evidence that takes, remains open.

2.4 Decisions with abstention and their operating characteristics

~~Three fields evaluate decisions that may abstain. In conformity assessment under measurement uncertainty, a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty and the specific and global risks of false acceptance and false rejection as the measures of a decision rule (ISO, 2017b; Joint Committee for Guides in Metrology, 2012). In analytical chemistry, the limit of detection is the smallest quantity a method detects with stated error rates (Currie, 1968, 1995), and in acceptance sampling and selection selection procedures and acceptance sampling the operating characteristic gives the probability of a decision as a function of against the true quality (Bechhofer, 1954). In machine learning, a with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970), summarised by risk coverage curves (Geifman and El-Yaniv, 2017), and interval forecasts are scored by their coverage and width (Gneiting and Raftery, 2007)(Chow, 1970; Geifman and El-Yaniv, 2017), learning to defer passes difficult cases to an expert (Madras et al, 2018), and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), also in design teams working with AI (Zhang et al, 2021). The decisive and safe distances are operating characteristics of this kind for, and the guard band of the procedure is the design-stage manufacturability decisions; what the framework adds is a unit set by production, which makes them comparable across characteristics and predictors, and their evaluation by evidence relation counterpart of a conformity-assessment guard band.~~

2.5 Production variation and measurement capability

Statistical quality control ~~separates the variation of a process into short-term compares short- and long-term components and compares its variation~~ with tolerances through capability ~~indices (Montgomery, 2019; Kane, 1986) and performance indices (Montgomery, 2019), with release criteria such as a performance index of at least 1.33 (Automotive Industry Action Group, 2006) and time-dependent models for drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and measurement systems analysis decides whether a gauge can resolve a tolerance (Automotive Industry Action Group, 2010; ?) separates~~

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Design margins (Eckert et al, 2019)	experience, analysis	buffer to the requirement	rarely	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy in physical units; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O’Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage; per decision possible on the measured item	yes
Conformity assessment, OC curves (ISO, 2017b; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks		yes
Selective classification (Chow, 1970; Geifman and El-Yaniv, 2017)	labelled data	class or reject; risk-coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; decisive and safe distance in floors by evidence relation	yes, per verdict	yes

repeatability from stability (Automotive Industry Action Group, 2010). The production floor is a ~~quantity of the same family, measured between batches of one design: it contains short-term scatter only through the batch average, and drift within a series in full.~~

repeatability-type limit of batch centres under within-run conditions; what is new is its use as the unit of a design decision rule. Table 1 places the framework among these approaches. ~~It: it~~ does not replace calibration or conformal prediction ~~; it uses them but scores them~~, as decision rules ~~and scores them~~, against production in a common unit, ~~which is what a design team needs to choose between them~~.

3 The framework

3.1 The decision problem and its scope

A design alternative a is a combination of product geometry and process settings that is produced as a series of parts. A quality characteristic c varies over the parts of a series. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet:

alternative a is *adequate* for c when the p -quantile of its parts satisfies $q_p(a, c) \leq R$.
~~The demonstration uses $p = 0.95$ and checks $p = 0.90$ and 0.99 ; requirements on a~~
~~lower limit~~ Lower or two-sided limits follow by a change of sign or by the absolute
deviation from the target; a tolerance of $\pm t$ on a wall angle about its design angle,
for example, is the limit $R = t$ on the absolute deviation. At the design stage q_p is
unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p
and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \quad (1)$$

The three verdicts mirror the decision rules of conformity assessment ~~, in which~~
~~uncertainty creates a zone where neither conformance nor non-conformance can be~~
~~proven (ISO, 2017b); here the uncertainty is that of the (ISO, 2017b), here for the~~
uncertainty of a design-stage prediction. A rule that returns a point ($L_r = U_r$) always
decides. The evidence consists of simulations of a and of the production record of other
alternatives, the *calibration alternatives*. ~~The decisions in scope are those of a design~~
~~team that has produced variants of a design family and has to decide on; it is used by~~
the process planner or tryout engineer for a further variant ~~, a new process setting or a~~
~~new family before producing it: the release of variants and process settings in process~~
~~planning and tryout, and the transfer of production knowledge to or process setting of~~
~~an existing tool, at tool tryout or before production part approval, and by the product~~
~~designer for a new family in platform and variant design.~~ The demonstration decides
on $p = 0.95$, which for a normal characteristic corresponds to a performance index of
about 0.55. Release criteria of series production, a performance index of 1.33 or more
(Automotive Industry Action Group, 2006), concern nonconforming fractions in parts
per million that a series of 500 parts cannot estimate without a parametric tail; the
framework reaches them through requirements anchored in capability (Section 5.8).

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts. ~~Each series is,~~ each cut into
batches of B consecutive parts, and each batch is summarised by a robust centre m
(here the 10 % trimmed mean). The production floor of characteristic c is

$$F_c = Q_\alpha \{ |m_i - m_j| : i < j \text{ batches of the same alternative} \}, \quad (2)$$

the α -quantile of the difference between two batches of one design, ~~pooled over~~
~~the produced alternatives (here $B = 50$).~~ Under a stationary normal model
with a standard deviation σ_b between batch means and σ_w between parts,
 $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2/B}$ at $\alpha = 0.95$ and series of ten batches; floors per geometry
~~are used:~~ the repeatability limit of precision experiments (ISO, 2025, 1994), applied
to batch centres under within-run conditions. The reference protocol of this paper
is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$, with
per-geometry floors where geometries differ). The floor rests on three assumptions:

batches of one design differ only by drift and scatter within a run (A1); the runs from which it is estimated are like those of future production (A2); and the ~~measurement adds little to it~~ measuring system is repeatable and its resolution small against the floor (A3, checked in Section 5.1). ~~It has three properties that~~

Four properties matter for its use. It has the unit of the characteristic, so ~~a margin expressed in floors does not depend on the unit or the origin of the characteristic~~. Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_b^2 + \sigma_w^2}/B$ at $\alpha = 0.95$, so the short-term capability enters through $\sigma_w/\sqrt{B}$ while drift within a run enters in full; the floor obtained after the parts of each series are randomly permuted contains scatter only, and its ratio to the floor measures the drift. ~~And the floor distances in floors are on a common scale for all characteristics and predictors, although equal distances do not mean equal nonconforming fractions, because the ratio of the floor to the scatter of the parts differs between characteristics (Table 3). It is a property of a production protocol: under drift it grows with the length of the series under drift, time between the batches compared, and it shrinks with B ; σ_b and σ_w are reported so that it can be recomputed for another protocol (Table 3). It contains whatever drifts between batches, the process and the measuring system alike, so without a reference artefact scanned through the series it is a production-and-measurement floor. And it~~ excludes the variation between runs, which ~~bounds it from below as a measure makes it a lower bound of the variation of series production. The~~ For a new family, whose floor is unknown before it is produced, ~~the floor of the produced family is used. Because the~~ floor resolves batch centres while the requirement concerns a quantile; ~~it is used as a unit, and,~~ the reproducibility of the quantile ~~itself~~ is reported alongside.

Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\text{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| / F_c \quad (3)$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative; ~~this is a practical threshold, not a test~~. For a continuous design or process parameter x with a local sensitivity $s_c = \partial \bar{m} / \partial x$, the minimum resolvable change is $F_c / |s_c|$: a change of x by less moves the centre of the characteristic by less than two batches of one design differ. It is local, a statement about centres; ~~and~~ a change below it can still move the tail of the distribution.

3.3 Decision rules

~~Twelve~~ Thirteen rules span what a design team can do with its evidence (Table 2); each is defined for any set of calibration alternatives. ~~Two use the simulation alone: the~~ and the first seven carry the results in the main text. The nominal simulation (M0) ~~and the worst case of a process-window simulation (M0w), as a robustness simulation is read in practice. Five combine the simulation with the production record. uses the simulation alone. The bias-corrected simulation (M1 adds to the nominal simulation)~~ adds the median offset between the calibration alternatives' q_p and their nominal simulation. ~~Me calibrates a parameter of the simulator instead: the friction coefficient of each lubrication pattern is chosen so that the simulated draw-in matches the produced~~

alternatives, in the sense of a calibration parameter of Kennedy and O'Hagan (2001) without a discrepancy model, and the median remaining offset is added; the scaled simulation (M1s) fits $q_p = a + b s_0$ to them by least squares, a linear multi-fidelity correction (Kennedy and O'Hagan, 2000) that rescales a simulated trend of the wrong size. The envelope rule (M2) takes the upper end of the simulation envelope over plausible incoming conditions, adds the median offset and a margin equal to the largest absolute residual of the calibration alternatives to the upper end of the simulation envelope over plausible incoming conditions. M5 models the offset between q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) over the alternatives' descriptors, so that the correction may vary with the design and process settings, and uses its with a 90 % predictive interval. M3 is a part-level model: gradient boosting (Friedman, 2001) learns the discrepancy between the measured characteristic and the matched simulation from the descriptors and the measured incoming conditions of the calibration parts, and the predicted parts of the new alternative give its q_p , with an offset and a margin from a leave-one-out over the calibration alternatives (a jackknife). Five rules use the production record alone and isolate what the simulation adds: the median q_p of the calibration alternatives (M1n), the same with the residual margin (M2n), the Gaussian process of, whose noise variance is bounded below by the variance between the produced series of one geometry and force, so that a new series of a produced setting is not predicted more precisely than produced series reproduce each other; M5n is the same process of q_p itself (M5n), the without the simulation. The nearest produced setting (NN) is a capability-database look-up that uses neither simulation nor learning. The six further rules include a friction-calibrated simulation (Kennedy and O'Hagan, 2001) (Mc) and part-level boosting model of the characteristic (M4), and the nearest produced setting (NN), which takes the mean q_p of the calibration alternatives of the design's family at the nearest process setting gradient boosting (Friedman, 2001) with a jackknife+ interval (Barber et al, 2021), with and without the simulation (M3, M3n). The nominal simulation and the envelope do not depend on the lubrication pattern, so a lubrication variant of a produced setting gives M0, M1, M1s, M2 and M5 no simulated information that distinguishes it from its siblings.

The margins are labelled by what they guarantee. With n calibration alternatives, the largest-residual margin of M2 and M2n covers a new exchangeable residual with probability $n/(n+1)$ when the offset is fixed and for a fixed offset and at least $(n-1)/(n+1)$ when it is estimated from the same for an estimated one, because the new residual is then covered whenever it falls within the range of the calibration residuals; it is the split-conformal quantile (Lei et al, 2018) at the highest level that n residuals can attain attainable level, not a 90 % interval. The Gaussian-process interval is model-based, and the jackknife margin of M3. In general, a distribution-free interval cannot certify a miss rate below $1/(n+1)$, so a miss rate of 5 % needs at least 19 produced alternatives in the relevant relation and one of 10 % needs nine. Under exchangeability the jackknife+ interval covers with probability at least 0.8 at the level of 0.9 used here (Barber et al, 2021), and the Gaussian-process interval is model-based. The level is a decision variable that trades decisiveness for safety, as the cost of errors would set the threshold of a reject option (Chow, 1970): 0.9 is used here, and 0.8 and

Table 2 The decision rules; the first seven carry the results in the main text

Rule	Prediction of q_p	Interval	Production evidence
M0 nominal simulation	simulation at nominal sheet thickness and friction	point	none
M1 bias-corrected simulation	M0 plus the median offset	point	q_p
M1s scaled simulation	$a + b$ M0, least squares on the calibration alternatives	point	q_p
M2 envelope + residual margin	upper end of the envelope plus the median offset	largest residual	q_p
M5 GP correction of the envelope	envelope plus a Gaussian-process offset over the descriptors	GP, 90 %	q_p
M5n GP of q_p	Gaussian process of q_p over the descriptors	GP, 90 %	q_p
NN nearest produced setting	mean q_p at the nearest produced process setting of the family	point	q_p
M0w process-window worst case	lowest to highest simulated value over the incoming conditions	range	none
Mc friction-calibrated simulation	simulation with friction calibrated on the draw-in, plus the median offset	point	centres, q_p
M3 simulation + boosting	matched simulation plus a boosting model of the discrepancy, part level	jackknife+	parts
M3n boosting, no simulation	boosting model of the characteristic, part level	jackknife+	parts
M1n median of the produced	median q_p of the calibration alternatives	point	q_p
M2n median + residual margin	M1n with the largest-residual margin	largest residual	q_p

Descriptors: geometry indicator, blank-holder force and lubrication rank (the order of the oil amount). ~~Largest residual: with n calibration alternatives, coverage $n/(n+1)$ at most.~~ GP: the noise variance ~~is~~ bounded below by the ~~sampling~~-variance between the series of one geometry and force. M3n is named M4 in the calibration alternatives' ~~q_p~~ released code; M6, a conformalised GP, appears only in the robustness analysis.

~~M4 carries no guarantee (Barber et al, 2021). Their empirical coverage is reported in Section 5.50.95 are reported in the robustness analysis.~~

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d F_c, \quad (4)$$

so that a is adequate exactly when $d \geq 0$. For a case (a, c) with interval $[L, U]$, let $u = (U - q_p)/F_c$ and $v = (q_p - L)/F_c$ be the distances by which the interval excludes the truth from above and below. At $d = +\delta$ the verdict is correct when $u \leq \delta$ and ~~wrong~~ (a false reject) when $-v > \delta$; at $d = -\delta$ it is correct when $v < \delta$ and ~~wrong~~ (a false accept) when $-u \geq \delta$. ~~Over a population of cases, let~~ Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$, ~~both~~

signs weighted equally and requirements below zero excluded for characteristics that cannot be negative; every case gives one verdict on an adequate alternative and, where $q_D - \delta F_c \geq 0$, one on a failing alternative, and every verdict has equal weight. As δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors (98 of 126 failing-side requirements remain feasible at 16 floors and 63 at 108). The *decisive distance* δ_d of a rule is the smallest δ beyond which $\kappa \geq 0.95$, and its *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$.

The two distances have five properties. They are exact: κ and ω change only where δ crosses one of the case's u , v , $-u$ or $-v$, so the distances are order statistics of these exclusion distances and need no grid; for a rule that always decides, $\delta_d = \delta_s$ is close to the 90 % quantile of its absolute prediction error in floors when no requirement is excluded. Always $\delta_s \leq \delta_d$, and between the two the rule sends designs to trial rather than misleading. They do not depend on the unit of the characteristic. They split into one-sided distances for failing alternatives, where the errors are false accepts, and for adequate ones, where they are false rejects, which matters when the two (false accepts) and adequate ones (false rejects) matter when the errors cost differently. And they are operating characteristics: functions of the true margin. The distances are points on operating characteristics over a population of design situations, like the operating characteristic of an acceptance plan (Bechhofer, 1954) or the risk coverage curve of a selective classifier (Geifman and El-Yaniv, 2017), not statements about a single verdict. A design team uses them before a requirement is fixed, to choose a rule for a family and evidence relation, to judge whether a requirement placed near the expected capability can be decided at the design stage at all, and to plan trials and production evidence. For a single verdict the observable quantity is the margin between the predicted interval and the requirement; the share of correct verdicts as a function of that margin is reported with the results.

In decision-theoretic terms, let a false accept cost c_{FA} , a false reject c_{FR} and a trial c_T , with correct verdicts free and trials error-free. Over the population at a true margin beyond δ_d , the expected cost of a verdict is at most $0.05 \max(c_{FA}, c_{FR}, c_T)$; and beyond δ_s at most $0.05 \max(c_{FA}, c_{FR}) + u(\delta) c_T$ with $u(\delta)$ the share sent to trial; the bound holds on average over adequate and failing alternatives weighted equally, and one-sided distances replace it when the base rates differ. Which rule to prefer then depends on is the analogue of the cost ratios; a worked comparison with realistic requirement margins is given in Section ??; indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95, and the pair plays the role of the acceptable and rejectable quality levels of an acceptance plan. A trial is itself not error-free: a short run reproduces the 95th percentile only to about one floor (Section 5.1), so no decision can be relied on closer to the requirement than the trial that would replace it.

3.5 Evidence relation, calibration budget and applicability guard

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced with

another variation that production may not resolve; a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); ~~and a new family~~ shares no produced alternative. ~~The two distances are evaluated for each relation~~ Beyond a factorial, the relation can be graded by a distance in descriptor space or by the effect-to-scatter ratio between the predicted characteristic and the nearest produced alternative (Section 6.5). The calibration budget is ~~their dependence the dependence of the distances~~ on the number k of produced alternatives, over all subsets of size k ; ~~and the calibration design asks which alternatives to produce first: subsets are classified by the relation of the new design to them~~ classified by relation. An applicability guard returns *trial* when the new design lies outside ~~what the calibration has seen~~ the calibration: its geometry is not ~~among the calibration alternatives~~ calibrated (family guard), a descriptor lies outside the calibrated range (range guard), or its simulated q_p lies outside the ~~range of the~~ calibration alternatives' simulated values (novelty guard); ~~all three use design stage information only~~, in the spirit of a valid input domain (Malak and Paredis, 2010).

3.6 Procedure

Figure 1 turns the framework into six steps. ~~Steps: steps 1 to 5 are carried out once per design family and, repeated when production evidence is added; step, and step 6 is carried out for every new design.~~ Step -1 frames the decision: ~~the family, its alternatives, the characteristics and the~~ (family, alternatives, characteristics, form of the requirement: ~~Step and conformance level~~); step 2 measures the floor ~~on the produced alternatives, and step with a stated protocol~~; step 3 states which differences between alternatives, and which local changes of a parameter, production can resolve; ~~a design question below the floor is not one the design stage needs to answer finer.~~ Step and step 4 calibrates the candidate rules, with and without the simulation, ~~and step.~~ Step 5 qualifies them by their ~~decisive and safe~~ distances for each evidence relation; ~~their calibration budget and the guard; when and fixes for each rule and relation a guard band g : the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts were correct, for requirements within 20 floors of the truth.~~ If the distances are too large for the requirements ~~expected~~ of coming designs, another variant is produced, ~~chosen by the calibration design.~~ Step, Step 6 ~~decides: it~~ establishes the relation of the new design to the produced evidence, applies the rule qualified for ~~that relation and the guard, and returns a verdict it and the guards, widens the rule's interval by g floors and returns meets or fails~~ with the margin ~~of the requirement~~ in floors, or a trial when ~~the guard fires or the margin lies below the rule's safe distance~~ a guard fires, no rule is qualified or the requirement lies inside the widened interval.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzlmann, 2026; Baum et al, 2026a). Square

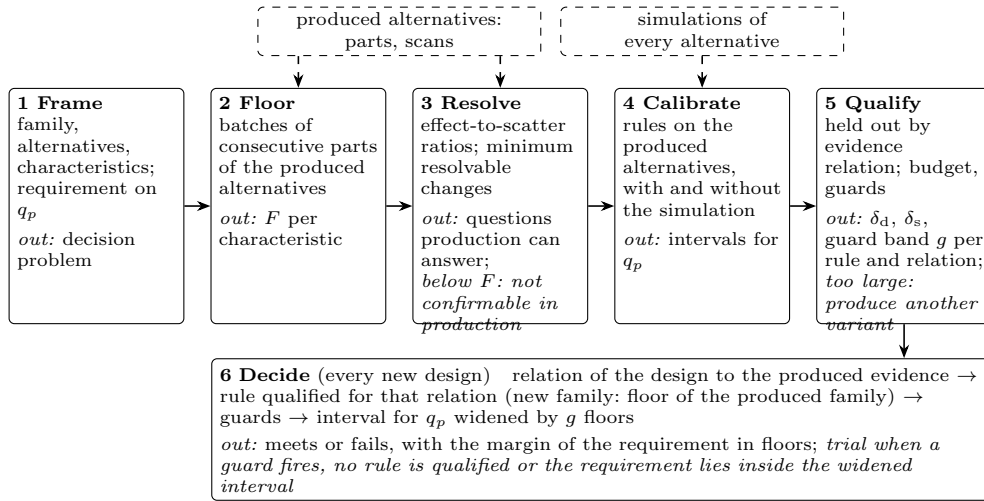


Fig. 1 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence it uses

blanks of DP600 ~~dual-phase~~ steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (~~operation~~ OP10) and cut (OP20) on one tool set with two geometries, concave and convex side walls, which differ in ~~the curvature of the base shape, the base curvature~~, wall angle and ~~the~~ bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine ~~oil application~~), with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m² of oil); ~~giving~~: 18 alternatives, each produced as one series of 500 consecutive parts. ~~Within a geometry the new designs are therefore new process settings and lubrication variants on an existing tool;~~ the change of geometry ~~is being~~ the only change of product design. For every part the dataset holds the press force, traverses of the sheet thickness and of the oil film before forming, ~~Every part has press signals, sheet and oil-film traverses~~ and 3D scans after ~~drawing and after cutting (0.077 both operations; 10 of the 9,000 parts carry no scan. The series warm up by 1.0–5.0 mm per pixel across the scan line, 0.158K over their first 150 parts from punch temperatures of 20.5–29.6 °C, their median sheet thickness ranges from 0.986 to 0.993 mm per line and 0.008, and the 100 kN series ran at 47 mm in height). Of the 9,000 parts, 10 carry no scan, leaving 8,990. /s against 78–79 mm/s. The documentation states neither the order and dates of the series nor whether the parts were scanned in production order.~~

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a): LS-DYNA ~~models of the same process on a quarter of the part, quarter models~~ with nodes about 1 mm apart, a scaled DP600 flow curve, ~~Coulomb friction with a constant coefficient between blank and tools, constant Coulomb friction~~ and spring-back after both operations; the ~~documentation does not state the yield criterion or the~~

yield criterion and unloading model, which govern springback (Yoshida and Uemori, 2002), are not documented, and the stroke speed is not represented. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm in six steps) and the friction coefficient (0.05–0.15 in eleven steps) at each geometry and blank-holder force. A part is matched to the simulation with the nearest sheet-thickness and the friction coefficient that the dataset’s linear mapping assigns to its measured-oil film oil film, an assumption of the dataset’s authors on which Mc, M3 and every matched simulation depend. The nominal simulation of an alternative is the one at 0.99 mm, the grid value nearest the measured median thickness of 0.990 mm median thickness, and friction 0.10, the middle of the range. The simulated characteristics carry extraction noise from the 1 mm node spacing; it is quantified in Section 5.2.

4.3 Quality characteristics and their measurement

Seven characteristics were extracted with the same definitions from the scans and from the simulated nodes, all referred to the plane of the cup bottom (Table ??): the mid-side and corner draw-in and the flange waviness, process indicators on the flange that the cut removes, and four characteristics a drawing would carry: the wall angle (design angle 20° concave, 10° convex), the cup depth, and the arm angle and bottom dome of the cut part, the last two with requirements on absolute values. The four sides of a part measure several of them twice, and this redundancy which exposes the measurement (Table ??). The flange width in the direction: the flange width along the scan lines scatters 4 to 18 times more than across them, uncorrelated with it and without serial correlation, and reads 2.3–3.5 mm larger; so the mid-side draw-in is therefore taken from the widths taken across the scan lines only. The north wall carries about 0.3° of uncorrelated scatter in every series and reads 0.9–1.6° low on the convex cups, and the south wall is shadowed; the wall angle is therefore the mean of the east and west walls, which also cancels a residual tilt of the part (on the convex cups the two are anti-correlated (the north wall carries uncorrelated scatter, the south wall is shadowed)). The corner mid-side draw-in uses the mean of the two flange diagonals, which differ by 1.6–2.2 is quantised in steps of 0.019 mm in every series, consistent with a small non-orthogonality of the scan axes to which their mean is insensitive. The quarter-model simulations are symmetric about the diagonal, so their values along both axes estimate the same quantity and are averaged. Differences that remain (0.38 floors) by the pixel pitch. Differences between scans and simulations—the simulations, such as the scanned surface against the shell mid-surface, edge treatment and spatial resolution—affect the absolute enter the offsets between the two but not the floors and the effects within production.

Quality characteristics, extracted with the same definitions from the scans and the simulated nodes

4.4 Evaluation design

Every produced alternative is judged as a new design under four schemes of grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant:*

the alternative alone is held out and the other eight ~~alternatives~~ of its geometry calibrate, ~~so two siblings at its blank-holder force, produced with another lubrication pattern, are force~~ among them. *New process setting:* all the three alternatives of one ~~blank-holder~~ force of a geometry are held out ~~together~~ and the six others calibrate; ~~the 300 kN alternatives are then interpolated and the is interpolated,~~ 100 and 500 kN ~~alternatives extrapolated in force extrapolated.~~ *New family:* ~~the alternatives of one geometry calibrate one geometry calibrates~~ the rules for the other ~~. The pooled (two transfers). Pooled~~ versions of the first two add the other geometry's alternatives to the calibration set. Every calibrated rule is evaluated with and without the simulation (M1/M1n, M2/M2n, M3/~~M4~~M3n, M5/M5n), so that the paired difference isolates what the simulation contributes, and the nearest produced setting (~~NN~~) is the baseline without simulation and ~~without~~ learning. A scope comprises 126 cases (18 alternatives, 7 characteristics) ~~and two transfer calibrations.~~

The distances are computed exactly ~~from the cases.~~ Their intervals come from 1000 resamples of the held-out alternatives within each geometry, ~~every resample taking one each with a~~ block-bootstrap replicate of ~~the every~~ true q_p of each alternative (batches of 50 parts resampled), so that the uncertainty of the reference enters; the resamples are, common to all rules, ~~which gives paired intervals for their differences.~~ These intervals ~~so that differences get paired intervals; they~~ hold the fitted rules fixed; ~~a.~~ A second bootstrap resamples ~~the alternatives of whole lubrication patterns within~~ each geometry, ~~which keeps every case in its evidence relation, and~~ refits every alternative-level rule ~~on each resample and repeats the evaluation, which adds the variability of the calibration sets~~ (Table A2).

4.5 Computation and development record

The constants of the floor and the requirement grid were ~~chosen fixed~~ before the decision results were seen ~~and are listed in Appendix ??;~~ as the repository history records ~~their dates.~~ The rules were not ~~all fixed in advance:~~ M0–M5 ~~and the three original calibration scopes were defined in a first analysis, the conformalised Gaussian process M6 was added as a candidate during it, and the rules were defined first;~~ M0w, Mc, M1n, NN, M2n ~~and~~ M5n, the grouped schemes, the exact estimator and the noise floor of the Gaussian processes ~~and the exact estimator of the distances were added in a second round of analysis, after the results of were added after the first had been seen. Every variant tried is reported, and first results; and a review of that version added M1s, the jackknife+ intervals, the comparisons between rules carry between-run noise floor and the analyses of Sections 5.4 and 5.7. The comparison of rules is therefore exploratory and carries~~ the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010); every variant tried is reported in the paper or the archived results. The pipeline ~~reads single files from the public archives by HTTP range requests, keeps only the extracted features, and~~ reproduces every result from the cached feature tables in about four hours on four ~~processor~~ cores.

5 Results

5.1 What the production floor contains

Table 3 gives the floor of each characteristic. Two In 95 % of batch pairs, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in, 0.046 mm in corner draw-in, and 0.078° in wall angle, 0.11° in arm angle, 0.011 mm in cup depth and 0.007 mm in bottom dome at 95 % confidence; resampling whole alternatives instead of batches, which respects the dependence of neighbouring batches under drift, widens the intervals most for the waviness (0.0021–0.0050 mm) and the arm angle (0.089–0.15°) (Table 3); one floor is 0.66 to 2.2 long-term standard deviations of the parts. The floors are set by drift: once within the run; without the time order of the parts is removed, they shrink by a factor of 2.1 (wall angle) to 4.5 (corner draw-in), and over a series the punch temperature rises by a median of 4.3 K (range, batches one position apart differ by 0.51–0.80 floors and nine apart by 1.1–6.4–1.5, and the normal model of Section 3.2 reproduces the floors to within 12 K), one of the sources of this drift.

Table ?? shows what the floor contains. Part-to-part noise that is independent between parts which includes the repeatability of the scanner would produce 15–33 % of each floor, so the floor is dominated by variation between batches, which a repeatable measurement cannot create. The redundancy of the scans had shown where the measurement did matter: with the flange widths along the scan lines and the north wall included, and the resolution is small against the floor except for the mid-side draw-in had a floor of 0.085 mm and a part standard deviation of 0.13 mm, of which the redefinition removes 40 % and 70 %; the wall angle kept its floor (0.081 against 0.078°) because the north wall's scatter averaged out within batches. Drift; assumption A3 holds for repeatability and resolution, but the stability of the scanner itself cannot be separated from drift of the process cannot be checked without a reference artefact scanned through the series. The floor also depends on the protocol: from the first 250 parts of every series it is 0.49–0.95 of the full value, and it varies by a factor of 0.6–2.2 over batch sizes of 25–100 and quantiles of 0.90–0.99 (Section 5.9). Nor do the process signals identify the source of the drift: batch centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20 , and removing the predicted drift changes the floors by -16 to $+12$ %. Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by up to 1.2 (waviness) to 9.8 floors (dome), and a floor taken across them is 1.5 to 9.1 times the reference. The decided statistic itself reproduces to about one floor: the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors at in 95 % confidence, and the block-bootstrap standard error of an alternative's q_{95} is 0.11 floors in the median (at most 0.39). Distances of pairs, so distances below about one floor are therefore at the limit of what production can confirm. Variation between runs is not in the floor: the three lubrication series of one geometry and force, which production mostly cannot tell apart, differ by a median of 0.5–1.6 floors and, at 95 %, by 1.2 (waviness) to 9.8 floors (dome), an upper bound of what separate runs of one design add beyond what production, or a trial, can confirm.

Table 3 Production floor per characteristic, pooled and per geometry, with its components

Characteristic	F	95 % interval	Concave	Convex	F/σ_{LT}	σ_b/F	σ_w/F	Drift
Draw-in, mid-side (mm)	0.051	0.045–0.061	0.050	0.052	1.2/1.5	0.32/0.34	0.91/0.66	3.3
Draw-in, corner (mm)	0.046	0.035–0.052	0.053	0.030	1.7/1.3	0.31/0.35	0.65/0.77	4.5
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0037	0.0021	2.1/0.8	0.33/0.30	0.55/1.31	2.5
Wall angle ($^\circ$)	0.078	0.066–0.10	0.081	0.078	1.6/0.7	0.32/0.26	0.68/1.39	2.1
Arm angle, cut ($^\circ$)	0.11	0.094–0.12	0.12	0.100	1.9/0.9	0.36/0.31	0.46/1.22	2.6
Cup depth (mm)	0.011	0.0096–0.012	0.0087	0.012	2.2/1.7	0.31/0.34	0.65/0.52	3.9
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0034	0.0082	1.4/1.5	0.33/0.36	0.76/0.55	3.9

Batches of 50 consecutive parts, 95 % quantile of the difference of batch centres. Intervals: interval from 2000 resamples of the batches within the alternatives, and of whole alternatives series. Ratios per geometry (clusters concave/convex). Part SD: σ_{LT} , median within-alternative long-term standard deviation of single the parts; σ_b , standard deviation between batch means corrected for $\sigma_w^2/50$; σ_w , within batches. Drift factor: floor over the floor of randomly permuted series. The decision analysis uses the per-geometry floors.

What the production floor contains: components and variants in floors

5.2 What production resolves and how far the simulation misses it

Figure ?? shows the 18 alternatives against the floor and Table ?? counts the contrasts that production resolves. All 63 geometry contrasts exceed the floor, with median effect-to-scatter ratios from 2.8 (waviness) to 129 (wall angle). Blank-holder and blank-holder force is resolved in 105 of 126 contrasts; the exception is the wall angle, which force barely moves (4 of 18). The lubrication pattern is resolved, but lubrication only in 51 of 126 contrasts and in at most 11 of 18 for any characteristic, so that for most characteristics an alternative's (42 with a bootstrap probability of at least 0.95), so a lubrication variant is often a near-replicate of it. The force sensitivity is local: between 100 and 300 kN, where the stroke speed changes as well, the minimum resolvable its siblings. The minimum resolvable force change is 21–67 kN for the cup depth, the waviness, the draw-ins and the arm angle, whereas between 300 and 500 kN, at equal speed, it is 43–105 between 100 and 300 kN for the depth, the draw-ins and the waviness and the arm angle no longer responds; most characteristics and lies beyond either force interval for the wall angle and the dome need 366–731 kN in either interval.

Single-factor contrasts that production resolves and minimum resolvable force change

Centre and 5–95 % range of each characteristic per alternative; the black bar is the production floor

⁰ σ_{ST} : standard deviation within subgroups of five consecutive parts; σ_{LT} : of the whole series (medians over series). White share: the floor that independent part-to-part noise alone would produce, $1.96\sqrt{2/B}$ times the successive difference standard deviation, over F ; it bounds the share a scanner's repeatability can explain. First half: floor of the first 250 parts of each series. q_{95} floor: 95 % quantile of the difference between the 95th percentiles of two batches of 100 parts. Between series: difference between the centres of the three lubrication series of one geometry and force, an upper bound of the variation between runs (lubrication effects included).

⁰ESR: effect-to-scatter ratio of two alternatives that differ in one factor; counts of contrasts with $ESR \geq 1$ and median ESR. MRC: floor over the sensitivity of the alternative centres to blank-holder force, between 100 and 300 kN (the stroke speed also changes) and between 300 and 500 kN (same speed); the sensitivity is local.

5.3 Simulation against production

The nominal simulation misses production by far more than a floor (Table ??). Its median offset reaches 109 floors for the cup depth of the concave cup and 108 for the corner draw-in of the convex cup, and within one geometry the offset varies by 8.5 to 39 floors between alternatives. In units, the simulation overestimates the corner draw-in by about 3 mm and the mid-side draw-in by 0.6–1.2 mm and underestimates the wall angle by 1.0–1.2°. Its blank-holder, its force effect on the draw-in is 2.6–2.7 to 5.1 times the measured one, which points to the friction model—a constant Coulomb coefficient where friction falls with contact pressure (Hol et al, 2012)—and to the modelled blank-holder contact rather than to the solver. The signed comparison exposes a sign error that the decision scale hides: and the cut arms of the concave cups bend upwards in every part and downwards in 89–100 % of the simulations ; so their absolute values agree better than their physics. The offset of the cup depth, –0.90 to –0.95 but upwards in every part. Part of the offset is definitional: the cup depth differs by a median 0.90 and 0.95 mm for all alternatives, is nearly constant and is consistent with a difference of reference—the shell mid-surface against, consistent with the scanned surface ; and the drawing depth of the model—rather than with forming physics. The simulated characteristics also carry extraction noise from the 1 mm node spacing (Table ??): 1.7–3.1 floors for the wall angle and 2.4–9.8 floors for the dome, which inflates the upper end of the simulation envelope of the dome by up to 21 against the shell mid-surface, and removing this offset alone brings the decisive distance of the nominal simulation from 108 to 65 floors.

Offset between the parts' 95th percentile and the nominal simulation, in floors

5.3 Decisions by evidence relation

Table 4 and Fig. 2 score the twelve rules. The nominal simulation (M0) needs requirements 108 floors from the truth (bootstrap interval 101–115) before 95 % of its verdicts are right, and ; it errs mostly by rejecting adequate alternatives : its safe distance is (safe distance 17 floors for false accepts but, 109 for false rejects (Table ??). Reading the process-window simulation by its worst case (M0w) makes it neither safe nor decisive (94 and 119 floors), because the window does not contain the simulation's bias.).

A new variant with a produced sibling. With the other eight alternatives of the family produced, every calibrated rule improves by an order of magnitude on the nominal simulation, and the simplest one is the most decisive: the nearest produced setting (NN), the mean q_{95} of the two siblings at the same blank-holder force, is right beyond 3.4 floors (2.0–4.5) and at 10 floors decides every case correctly. The Gaussian process without the simulation (M5n) decides at 6.1 floors and is safe at 1.0, and with the simulation (M5) at 6.7 and 1.4. The interval rules are safe at or below 1.4 floors, but the envelope with the residual margin (M2), 0.17 mm of mid-side draw-in. The scaled simulation follows at 4.9 floors; the Gaussian processes decide at 7.2 floors and

⁰Offset: median over the nine alternatives of a geometry of $(q_{95} - s_0)/F$; s_0 the nominal simulation, on the decision scale (absolute values for the arm angle and the dome); spread: its range over the alternatives. Matched, signed: median difference between the centre of the parts and of their matched simulations, in the characteristic's unit, with signs; the simulated arm angle of the concave cup has the opposite sign of the parts'.

are safe below one floor; the envelope rule is safe at 0.15 floors but sends a third of the designs to trial at 10 floors and ~~is decisive only at 23 floors, decides only at 23~~. Bias-correcting the nominal simulation (M1, 15 floors) is worse than ~~taking~~ the median of the produced alternatives without it (M1n, 9.1), ~~and calibrating the friction coefficient (Mc, 27 floors) is worse still~~.

A new process setting. ~~When~~ With every alternative at the new design's blank-holder force ~~is force~~ held out, all rules lose. The nearest produced setting is right beyond 8.5 floors (5.5–10.2) ~~and the median of the produced alternatives beyond 10~~; the interval rules ~~are decisive only at 13–18 floors and safe at 5.4–8.1 (M4 13/7.9, M2n 15/decide only at 13–31 floors, are safe at 5.9, M2 18/8.1, M5 18/7.4, M5n 18/5.4; M3 32/16),~~ 16 and at 10 floors ~~they send a fifth to a third send 18–37 %~~ of the designs to trial. Interpolating in force ~~(300 kN from 100 and 500 kN)~~ is much easier than extrapolating: the nearest setting decides at 4.7 against 9.5 floors, and M2 and M5 are safe at 2.7 and ~~2.9–2.8~~ against 9.9 and 8.9 floors. The scaled simulation decides at 19 floors and, extrapolating a fitted trend, falsely accepts failing alternatives as far as 164 floors from their limit.

A new design family. ~~Calibrated only on the other geometry, In the two transfers~~ every rule without the simulation fails (131–133 floors), because nothing in one family's production record describes the other. With the simulation, the bias correction decides at 35 floors, M5 at 38 and M3 at ~~48~~49, and the envelope ~~with its residual margin rule~~ is the safest ~~rule~~ at 16 floors; no rule is safe closer than that. ~~The simulation thus reduces the distance at which a new family can be decided from about 130 floors to 35–51, but not to a distance at~~ Scored in the floor of the produced family, which a design team would ~~rely on it near the requirement~~ have, M5 decides at 35 and is safe at 31 floors and the envelope rule at 41 and 26. Sixteen floors are about 0.8 mm of mid-side draw-in or 1.3° of wall angle.

5.4 Where the distances come from

Three features of the evaluation design shape the distances. First, the sibling relation mixes replication with variation: in 58 of the 126 new-variant cases neither sibling differs resolvably from the held-out alternative, and the nearest produced setting is then right beyond 0.85 floors, the reproducibility of q_{95} between near-replicate runs; when both differ resolvably (34 cases), it needs 4.9 floors and M5 9.7. Second, the extrapolation cases are confounded with stroke speed: the production-only rules lose almost only at 100 kN, where the press also ran slower (the nearest setting decides at 11 floors against 4.9 at 500 kN), whereas the simulation-based interval rules are unsafe at both ends. Third, the production record of the other family does not help within a family: pooling it makes most rules less decisive (M5 from 7.2 to 11 floors for a new variant, M2 from 23 to 53, M2n from 19 to 325), although it raises the coverage of M5 from 86 to 90 % and makes the centre of its interval the most accurate point prediction within the family (2.9 floors against 3.4 for the nearest setting).

The relation therefore does not govern decision fitness in general. Over the eleven calibrated rules and three relations, the relation explains 69 % of the variation of the log decisive distance and the rule 6 %, but this share comes from the new family, where the rules without the simulation fail by construction; within a family the rule explains

Table 4 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

Rule	New variant		New process setting				New family		Pooled	
	δ_d/δ_s	FA	δ_d/δ_s	FA	interpolated	extrapolated	δ_d/δ_s	FA	variant	setting
M0	108	17	108	17	109	106	108	17	108	108
M1	15	15	24	24	11	25	35	33	27	28
M1s	4.9	4.7	19	164	19	19	122	122	15	20
interval	4.7–6.3		14–21		19–20	15–19	121–123			
M2	23/0.15	0.30	18/8.1	8.3	18/2.7	18/9.9	51/16	11	53/0	53/1.0
M5	7.2/0.75	0.15	19/6.8	7.6	19/2.8	19/8.9	38/28	32	11/0.05	28/5.4
interval	6.6–8.4		18–20		18–20	18–20	35–39			
M5n	7.2/0.55	0.10	17/6.1	9.0	16/0	18/8.6	132/130	167	9.8/0.20	16/2.7
interval	6.4–7.5		16–18		15–16	16–19	130–133			
NN	3.4	2.8	8.5	10	4.7	9.5	131	169	3.4	8.5
interval	2.0–4.5		5.5–10		4.3–5.1	6.6–11	129–131			
M0w	119/94	9.2	119/94	9.2	119/94	119/77	119/94	9.2	119/94	119/94
Mc	27	28	39	39	14	40	38	42	32	33
M3	23/0	0	31/16	13	29/13	33/17	49/21	20	25/0	32/5.1
M1n	9.1	10	10	12	4.8	12	131	173	130	130
M2n	19/0	0.20	15/5.9	9.2	13/0	16/9.1	132/129	165	325/0	325/4.0
M3n	11/0.05	0	13/7.4	9.2	15/7.0	13/7.8	133/128	164	15/0	15/4.9
M6	17/0.05	0	56/6.2	5.8	47/0	56/8.1	41/25	29	–	–

Decisive/safe distance δ_d/δ_s over δ_d/δ_s all seven characteristics; one number for rules that always decide ($\delta_d = \delta_s$). New variant FA: a sibling at safe distance on the same blank-holder force is produced false-accept side (leave one alternative out within the family failing alternatives); new setting, Interval: all alternatives bootstrap 95% interval of one force are held out the decisive distance (interpolation at 300 kN, extrapolation at 100 or 500 kN fits fixed); new family: calibration on the other geometry. 126 cases per rule relation (42 for interpolation interpolated, 84 for extrapolation extrapolated); a new family rests on two transfers. Intervals Pooled: the other geometry's alternatives added to the calibration set. Below the line, the further rules; M6, the M5 mean with a conformal margin on standardised leave-one-out residuals, appears only in Table ?? the robustness analysis.

68% and the relation 15%. The best attainable distance still grows with the novelty of the design, from 3.4 floors with a sibling to 8.5 for a new setting and 35 for a new family.

5.5 What the simulation and learning contribute

Pairing each calibrated rule with its counterpart without the simulation isolates what the simulation adds (Table ??). Within a family as it is used. Added as an offset, it adds nothing : the decisive distance of within a family and degrades two rules: M5 differs from that of decides where M5n by +0.6 floors (95% interval does (difference 0.0 to +1.6), that of floors, -0.4 to +1.1), and M2 from M2n by +3.9 (+0.4 and M1 decide 3.9 (+0.5 to +7.4) ; and that of M1 from M1n by +6.2 and 6.2 floors (+4.6 to +9.7) ; for a new process setting the bias correction loses 13.5 floors by using the simulation further out than M2n and M1n, because the simulated force effect is too large. Rescaled, it adds information within a family, where the scaled simulation decides 4.2 floors closer than the median of the produced alternatives (-5.4 to -0.5), though 1.5 floors less close than the nearest setting; for a new setting the rescaled

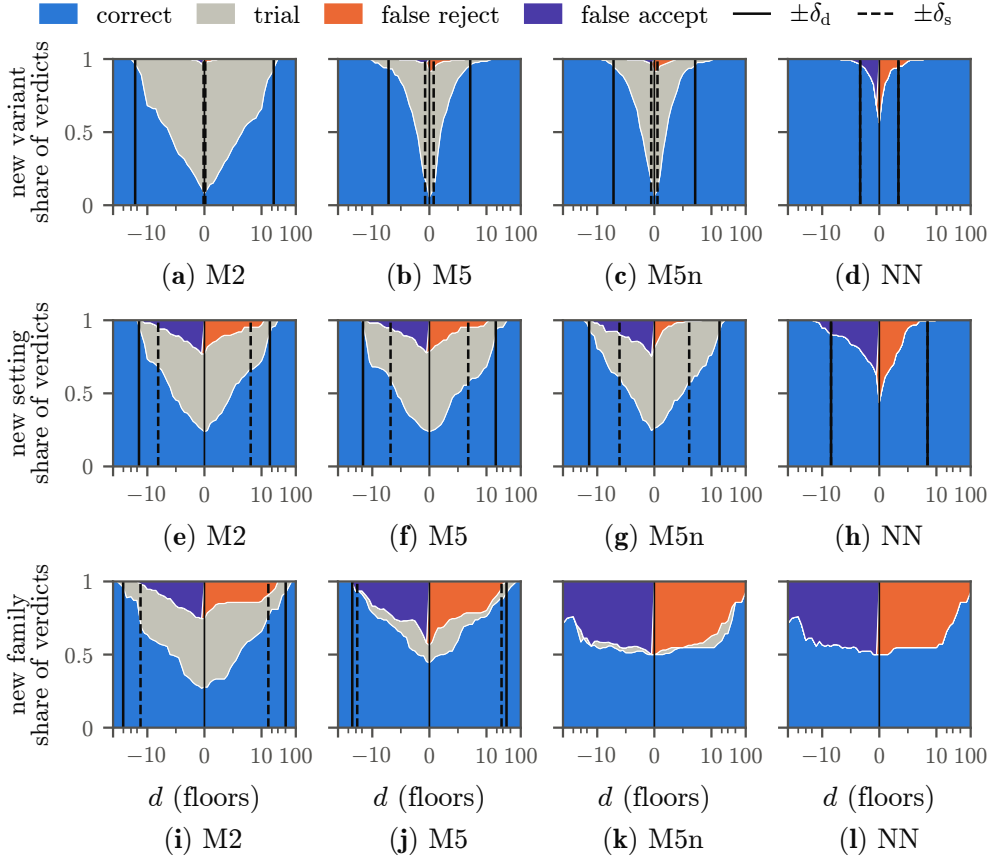


Fig. 2 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ± 10 floors; 126 cases per point panel) for four rules with the decisive and three evidence relations safe distances marked

trend is worse than none. Only for a new family is the simulation decisive, by about 93 floors. Learning adds little over the nearest produced setting: within the family 94 floors, and even there production knowledge carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family's deviation from its own design angle decides at 2.0 floors, against 7.3 for M5 is 3.3 floors less decisive than NN (+2.1 to +5.1) and 2 floors safer, and for a new setting 9.4 floors less decisive. The ranking between learners rests on their. A surrogate of the simulations reproduces them to within 1.8 floors for 12 of the 14 characteristics and geometries, yet its decision fitness is that of the simulation it reproduces.

Learning adds intervals rather than on their predictions: used accuracy. Used as point rules, the centres of the M5, M5n and M4-M3n intervals decide at 3.4, 2.8 and 4.8-3.3, 3.1 and 3.7 floors within the family, close to the nearest produced setting (Table ??). The part-level models keep their ranking under other learners

and settings, including quantile-loss boosting and settings selected by an inner leave-one-out (Table ??).

The intervals are calibrated only where the calibration alternatives are exchangeable with the new design (Table ??). Within the family the residual margin of M2 covers the true q_{95} in 88.9% of the cases—exactly the $n/(n+1) = 8/9$ that eight residuals attain—and the part-level jackknife of M3 in 93%; as close as the nearest setting, so the ranking of the learned rules rests on the width of their intervals. With the noise floor set by the variation between runs, the Gaussian-process intervals cover 71–85–86% although their nominal level is of the true q_{95} within the family against a nominal 90%, and they are narrow (1.1–1.2 floors in half-width) because the fits interpolate: the noise level sits at its lower bound in 35–38% the largest-residual margin 89% of the fits and a length scale at a bound in 28–48% (it guarantees at least $(n-1)/(n+1) = 7/9$ under exchangeability, which these cases lack) and the jackknife interval of M3 96% (Table ??). For a new setting every calibrated interval covers only 39–56% the calibrated intervals cover 43–55%, and for a new family the Gaussian processes cover none, M2 47% and M3 40% at most 12%. What a designer observes for a single verdict is the margin between the predicted interval and the inductive biases explain much of this: trees do not extrapolate, two force levels identify at most a linear trend, and in the median new-setting fit the length scale of the requirement. With requirements spread over the 20 floors around the truth, the share of correct verdicts among the decided ones grows with that margin for a new variant and a new setting—for the nearest produced setting from 86% at any margin to 95% beyond 5 floors and 99% beyond 10 for a new setting—but not for a new family, where M5 is right in 72–75% of its decided verdicts whatever the margin. The observed margin is a guide to a verdict only where the rule is calibrated for the evidence relation: force sits at its lower bound, so the Gaussian process reverts to its mean.

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1087 5.6 Calibration budget and calibration design

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Figure 3 and Table ?? show how the distances depend on the number of produced alternatives of the family and on how their relation to the new design relates to them. A single produced sibling decides: with one alternative at the new design's force, the bias correction, the residual-margin envelope rule and the nearest setting all coincide by construction (the simulation terms cancel and a single residual gives no margin) and decide at 3.5 floors (2.3–4.6) without a wrong verdict at 10 floors; and further. Further alternatives do not improve the nearest setting. They; they make the bias correction worse (12–18 floors), because the offsets of other forces dilute the sibling's the envelope rule safer but less decisive, and the interval rules safer but not more decisive: M2 becomes safe at 0.2–2.2 floors and decisive at 15–23 Gaussian processes safer and more decisive (from 12–14 floors with two alternatives to 7.2 with eight). Without a sibling, bracketing the new force buys safety but not decisiveness: with produced alternatives on both sides, M2, M5 and M5n from two alternatives on the interval rules are safe at 0.5–40–4.8 floors; M2 and M5n are never wrong with no wrong verdict at 10 floors and M5 in under 0.5%, but they but decide only at 15–25 floors, while the nearest setting decides at against 4.7–6.0 floors for the nearest setting.

1104

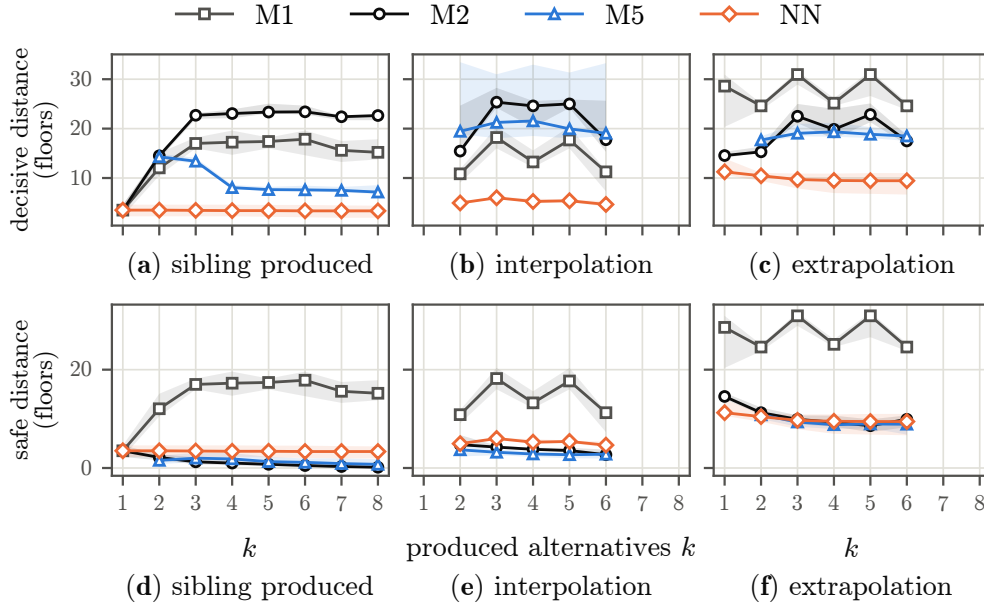


Fig. 3 Decisive and safe distances against the number of produced alternatives of the family, by the relation of the new design to them; bands: 95 % intervals over the held-out alternatives

Extrapolation stays unsafe at every budget: ~~no rule is safe closer than 8.5 floors, and at 10 floors 2–13 % of the verdicts of the interval rules and the nearest setting are wrong, and 26–28 % of those of the bias correction.~~ Averaged over all subsets, the distances shrink with the ~~number of produced alternatives~~ budget mainly because the share of subsets ~~that contain~~ containing a sibling grows, from 46 % ~~of the cases~~ with two alternatives to 96 % with six; ~~what a produced alternative is worth depends on its relation to the designs to come~~ the odd-even pattern of the bias correction comes from its median offset, which averages two forces only for an even number of alternatives. With calibration series of 50, 100 or 250 parts instead of 500, no distance within a family moves by more than 1.2 floors, and for a new family by at most 4 floors (Table A2): the qualification needs short runs of several variants, and only the floor needs a run long enough to show the drift.

~~Calibration design: distances and wrong verdicts at 10 floors by the relation of the new design to the produced alternatives~~

5.7 ~~Applicability guard~~ The procedure's decision rule

~~A guard is useful only where it can fire, which within this design means a new process setting and a new family (Table ??). For~~ What a designer observes for a single verdict is the margin between the predicted interval and the requirement, and its value as evidence depends on the relation (Fig. 4): for a new variant at least 95 % of the decided

⁰ All subsets of k produced alternatives of the family. Sibling: the subset holds an alternative at the new design's force; interpolation: no sibling, forces on both sides; extrapolation: no sibling, the new force outside. Wrong: share of wrong verdicts at $|d| = 10$ (%). M5 from $k = 2$; interpolation needs at least two alternatives.

1151 verdicts of the interval rules are correct at every margin; for a new setting the ~~range~~
 1152 ~~guard refuses exactly the extrapolated forces, the 84 of 126 cases whose distances are~~
 1153 ~~largest, and among the interpolated cases it lets through, M2, share grows with the~~
 1154 ~~margin, for the nearest setting from 85 % to 95 % beyond 5 floors; for a new family~~
 1155 ~~M5 and is right in 71–74 % of its decided verdicts whatever the margin. Step 5 turns~~
 1156 ~~this into guard bands: none for the interval rules within the family and 0.25 floors~~
 1157 ~~for the nearest setting; 2.0 floors for an interpolated setting, with which the nearest~~
 1158 ~~setting are never wrong decides at 6.7 floors, is safe at 2.7 and sends no design to~~
 1159 ~~trial at 10 floors; 7.0 floors for an extrapolated one, with which it decides at 17 floors~~
 1160 ~~and sends a quarter of the designs to trial at 10 floors. The novelty guard, which~~
 1161 ~~compares the simulated q_{95} of the new design with the calibration range, also refuses~~
 1162 ~~84 cases but not the same ones; among the 80 verdicts it lets through,;~~ and for a new
 1163 family no guard band up to 20 floors reaches 95 %, so every design goes to trial. The
 1164 guard bands are chosen on the evaluation data, which makes the trial rates optimistic.
 1165 For a new setting the range guard refuses exactly the extrapolated forces, after which
 1166 ~~M2 and, M5 are wrong twice and the nearest setting never, but 72–96 % of the verdicts~~
 1167 ~~it refuses make no wrong verdict at 10 floors; the novelty guard also refuses many~~
 1168 ~~cases that would have been correct. For a new family the family and range guards~~
 1169 ~~refuse everything. The novelty guard lets 12 of 126 cases through, all of them decided~~
 1170 ~~correctly.~~

1171 *A worked example.* A tryout engineer has to decide whether the concave cup at
 1172 300 kN with medium lubrication, a new process setting, can be released without a
 1173 trial run, with the six alternatives at 100 and 500 kN produced and a requirement
 1174 that 95 % of the parts have a mid-side draw-in or waviness, and among their 24
 1175 verdicts at 10 floors M5 is wrong in 6 (25 of at most 15.30 %) and M1 in 6, whereas
 1176 mm. The floor is 0.050 mm, the relation is interpolation and no guard fires. The
 1177 nominal simulation predicts 15.66 mm and fails the design. The nearest produced
 1178 setting predicts 14.99 mm, a margin of 6.2 floors, more than its guard band of 2.0
 1179 floors, so step 6 returns meets; beyond a margin of 6 floors every decided verdict of
 1180 this rule for an interpolated setting was correct. ~~M2 is never wrong and sends 12, M5~~
 1181 ~~and M5n return intervals reaching 15.37, 15.58 and 15.33 mm and send the design to~~
 1182 ~~trial. A guard on the simulated value therefore protects a rule only as far as the rule's~~
 1183 ~~own interval does~~ The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below the
 1184 limit: the procedure was right, the simulation wrong and the interval rules cautious.
 1185

1186 5.8 Requirements ~~from general~~ anchored in capability and in 1187 tolerances

1188
 1189 Requirements from the angular classes of ISO 2768-1 (ISO, 1989), applied to the wall
 1190 angle against its design angle and to the arm angle against flat with the row for the
 1191 shorter leg of each, turn the verdicts into concrete design decisions (Table ??). Of the
 1192 108 decisions, 69 concern an alternative that fails its limit, and the limit lies within
 1193 5 floors of the truth in 32 and within 10 floors in 55; deciding the worst side of a
 1194 part instead of the mean of its sides would change the truth in 12 of them, since a
 1195 general tolerance applies to every feature. For a new variant the nearest produced
 1196 setting decides all 108 correctly and M5 93 Upper limits at which an alternative has

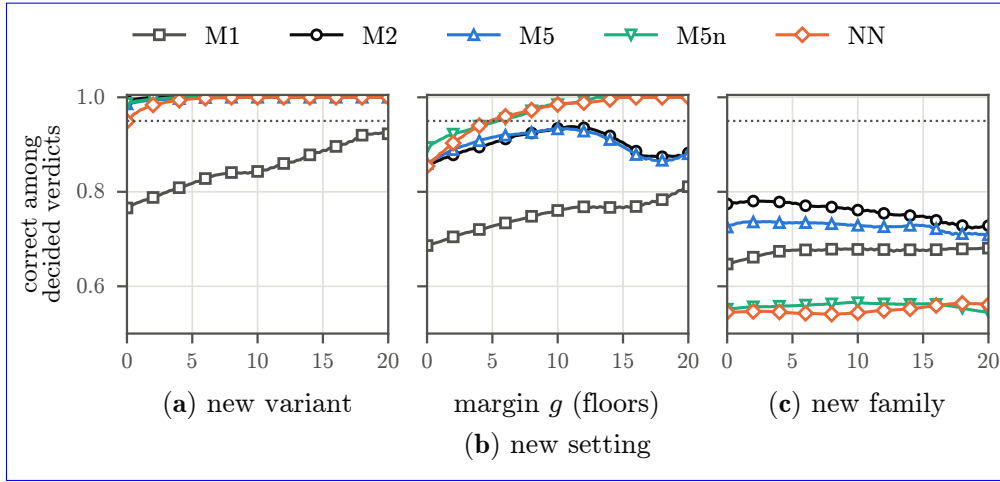


Fig. 4 Applicability guards: distances and wrong Share of correct verdicts among the cases a guard lets through decided ones against the observable margin between interval and requirement, for requirements within 20 floors of the truth

a performance index of 1.0, 1.33, 1.67 or 2.0 lie a median 1.0, 1.7, 2.5 and 3.1 floors above its 95th percentile, inside every decisive distance; every alternative meets them, so a verdict is right, a false reject or a trial. At an index of 1.33 the nearest setting still says meets in 91 %, with no false accept; within 5 floors of the limit, of the new-variant cases, because most of its errors are far smaller than its 95 % distance, whereas M5 is right in 75 % says so in 44 % and sends the rest to trial, M2 is right in 25 %, and the bias-corrected simulation accepts three of the six failing alternatives there. For a new process setting the nearest setting is right in 97 % and the Gaussian process without the simulation in 92 %; for a new setting the two are right in 65 and 32 %, whereas M5 and M2 send 22–25 % to trial and falsely accept 4 % of the failing alternatives. For a new family M5 is right in 68 % but rejects 46 % no rule exceeds 54 % of the adequate alternatives and accepts 9 % of the failing ones; M2 never accepts a failing alternative but sends 39 % to trial. Release-level requirements are therefore decided at the design stage only within a family, and then by the production record. General angular tolerances of ISO 2768-1 (ISO, 1989) give a coarser scenario that mostly tests whether a rule recognises gross failures (32 of its 108 decisions lie within 5 floors of the limit). With a false reject costing half at half the cost of a false accept, the cheapest rule per decision its cheapest rule is the nearest produced setting setting at any cost for a new variant at any trial cost, the interval rules without the simulation (M2n, M4 and M5n, tied); for a new setting the production-only interval rules while a trial costs at most a tenth of a false accept, and the nearest setting from a fifth, and beyond; for a new family M2 up to a tenth and M5 from a fifth. A drawing of a deep-drawn part would normally carry profile and position tolerances to datums (ISO, 2017a, 2021), which the data do not allow to evaluate; the general tolerances serve as requirements with realistic margin trial of every design while a trial costs at most a tenth, the envelope rule at a fifth and M5 beyond.

5.9 Robustness

Table A2 repeats the analysis The analysis was repeated under variants of the data, the constants and the Gaussian-process specification. The variants rescale the distances but leave the conclusions in place: in every variant that includes it, the nearest produced setting remains the most decisive rule for a new variant and a new setting and M2 the safest for a new family, and the rules without the simulation fail for a new family (91 floors or more); only rules within a few floors of each other change places, such as M1, M5 and Me for a new family. Correcting every characteristic for the model specifications (Table A2). Adjusting for punch temperature changes the pooled floors by -14 to $+4\%$ and the distances of the calibrated rules by at most no calibrated distance by more than 2.6 floors. Smoothing the simulation removes its extraction noise but not its bias: M5 is unchanged within the family and the envelope rule becomes less safe. Dropping the warm-up of the first 150 parts shrinks the floors and so lengthens the distances (the nearest setting 4.1 floors with a sibling, 12 for a new family (25 instead of 16 floors). Deciding on the 90th or 99th percentile instead of the 95th moves no distance by more than 1.7 floors. Floors from batches of 25 or 100 parts rescale the distances by factors of 0.80–0.96 and 0.97–1.16 (setting), and floors at the 90% or 99% quantile by 1.16–1.46 and 0.54–0.83. Requiring 90% or 99% correct verdicts instead of 95% shortens the decisive distances by up to 43% (within the family) or lengthens them by up to 93% (for a new setting), as an operating characteristic should. The taken between series, 1.5 to 9.1 times the reference, shorten them (0.75 and 2.7 floors). Other floor protocols rescale the distances by 0.60 to 1.42, the Gaussian-process variants—a Matérn kernel, a linear trend, one-hot lubrication, a floor on the length scales and the first specification without the noise floor—keep M5 at 6.4–6.9 floors for a new variant and 17.7–18.9 for a new setting. Leaving out one lubrication pattern behaves like a new variant (nearest setting 3.4, M5 6.4/1.1 floors), because the siblings at the same force remain. Refitting every rule on resampled calibration sets specifications move them by at most 2.1 floors, and in the unit of the reproducibility of q_{95} the nearest setting decides at 2.8 and 6.9 floors. In every variant that includes it, the nearest setting remains the most decisive rule within a family and the envelope rule the safest for a new family. The refitting bootstrap widens the intervals, most for the Gaussian processes of the learned rules most (M5 4.0–11 floors for a new variant; 5.0–18.6 floors, against 2.2–7.0 for the nearest setting), whose fits on calibration sets with repeated alternatives are unstable. Replacing each part's oil film by its pattern's median makes M3 less decisive (33 instead of 17 floors), and a conformalised Gaussian process (M6), the M5 mean with a margin on standardised leave-one-out residuals, is safe at 0.4 floors within the family but decisive only at 39: it buys safety for 2.3–4.6), yet the nearest setting is the over-confident Gaussian-process

⁰Wall and arm angle, classes f/m, c and v, 18 alternatives: 108 decisions per rule and scope, 69 with a failing alternative and 32 with the limit within 5 floors of the truth, right: correct verdicts; <5: correct verdicts among the decisions within 5 floors; FA: false accepts among the failing alternatives; trial: sent to trial.

~~intervals at the cost of decisiveness~~ more decisive in all 200 resamples, against M5 and M5n and for a new setting alike.

~~Decisive/safe distances (floors) under the robustness variants~~

6 Discussion

6.1 What the evaluation establishes

Five findings answer the ~~three research questions~~ beyond the particular numbers. First (RQ1), production has a resolution, and the floor makes decision fitness comparable across characteristics and predictors: it is set by drift within a run rather than by the scatter of single parts or by the measurement, the decided percentile reproduces to about one floor, and the distances of very different rules can be read on one scale. ~~It research questions. Two follow from the definitions and hold wherever the framework is applied: the floor puts every rule and characteristic on a common production-referenced scale, though not on a common scale of nonconforming fractions, and as a within-run quantity it is a lower bound for series production, since separate runs of one design may differ by several floors more.~~ Second (RQ2), the relation of a new design to the produced evidence governs decision fitness more than the model does. With a produced sibling, the best rules calibrated on production decide within a few floors; for a new process setting no rule decides closer than 8.5 floors, and extrapolating in force doubles the decisive distance of the nearest produced setting and more than triples the safe distances of M2 and M5; for a new family no rule is safe closer than 16 floors. Third, ~~of the variation of series production (RQ1); and an interval rule trades decisiveness for safety, which the two distances make explicit, and reaches its nominal level only where the calibration alternatives are exchangeable with the new design.~~ Three are observations on this dataset and hypotheses for confirmation: the best attainable distance grows with the novelty of the design, while within a family the choice of rule matters more than the relation (RQ2); production evidence buys safety before decisiveness⁰, since interval rules become safe with a few produced alternatives that bracket the new design but remain indecisive, and only and decisive only with near-replicates make them decisive (Section 5.6). Fourth, interval rules trade abstention for safety, the two distances make the trade explicit, and the nominal level of an interval is reached only where the calibration alternatives are exchangeable with the new design. Fifth (RQ3), in this data regime the simulation's contribution is confined to a new family; and with this simulator and six to nine produced settings per fit, the simulation contributed as an offset only across families, a fitted scale made it useful within a family but harmful for a new setting, and learned models ~~add-calibrated abstention rather than accuracy over-predicted as well as~~ the nearest produced setting while adding abstention (RQ3).

⁰ Alternative-level rules refitted for every variant; empty cells: the variant does not affect the rule. Smoothed simulation: nominal value and envelope from a quadratic response surface over thickness and friction. Floor variants rescale the distances with other floors. Leave one pattern out: columns M1, M2, M5 and NN of the first block. Refit bootstrap: 95 % interval of the decisive distance when the alternatives of each geometry are resampled and every rule is refitted (200 resamples).

1335 **Table 5** Guidance for design-stage manufacturability decisions, with its basis

1336 Situation	1337 Guidance	1338 Basis
1338 <i>From the framework (general)</i>		
1339 Before trusting a predictor	1340 Measure the floor on consecutive production of the family with a stated protocol and report σ_b , σ_w and its lag dependence; treat it as a lower bound of series variation.	1341 definitions
1342 Qualifying a rule	1343 Evaluate it for the evidence relation of the coming designs by grouped cross-validation, and split near-replicate siblings from resolvable ones.	1344 definitions
1345 Choosing a rule	1346 Compare it with the nearest produced setting and with its own version without the simulation, on decisive and safe distance jointly or by expected cost.	1347 definitions
1348 Deciding a single design	1349 Apply the rule's guard band to the observable margin; a margin inside the guard band, or a fired guard, means a trial.	1350 definitions
1351 Fixing a requirement	1352 If the expected margin to the capability is below the safe distance of the best qualified rule, plan a trial; no decision is reliable closer than a trial reproduces the quantile.	1353 definitions
1354 <i>Observed on this dataset (18 alternatives, 2 geometries, 7 characteristics)</i>		
1355 New variant with a	1356 nearest produced setting right beyond 3.4 floors (0.85 for a near-replicate); calibrated interval rules safe below one floor	1357 126 cases
1358 produced sibling	1359 nearest produced setting right beyond 4.7 floors	1360 42 + 84 cases
1361 New process setting	1362 interpolated and 9.5 extrapolated; interval rules safe at 5.9–16 floors	1363
1364 New family	1365 best rule safe at 16 floors; rules without the simulation fail; a transferable drawing nominal helps	1366 two transfers
1367 Release-level requirements	1368 lie within 1–3 floors of the 95th percentile, inside every decisive distance	1369 126 cases

1359 6.2 Using the framework in design

1361 Table 5 separates guidance that follows from the framework ~~and plausibly holds~~
1362 ~~elsewhere~~ from observations on this dataset, ~~with their evidence base. The general~~
1363 ~~guidance turns the distances into planning questions.~~ Before a requirement is fixed, the
1364 expected margin between ~~the requirement and the capability of the design requirement~~
1365 ~~and capability~~ can be compared with the safe distance of the best rule qualified for the
1366 ~~evidence relation of that relation of the~~ design; if the margin is smaller, ~~no design-stage~~
1367 ~~decision can be trusted and~~ a trial or further production evidence has to be planned.
1368 Release-level requirements make this the usual case: they lie within a few floors of the
1369 capability, where only a near-replicate production record decides reliably, consistent
1370 with the practice of requiring a production trial run for a process change. For a single
1371 ~~verdict, the margin between the predicted interval and the requirement is what a~~
1372 ~~designer observes, and it is informative only where the rule is calibrated for the relation~~
1373 ~~(Section 5.5)~~ design, step 6 gives a verdict whose guard band a designer can see, and
1374 ~~the worked example shows that the most decisive rule is also the most transparent:~~
1375 ~~the nearest produced setting points to the produced alternative it copies, whereas a~~
1376 ~~Gaussian-process interval does not.~~ Which rule to prefer ~~then~~ depends on the costs:
1377 ~~with the realistic margins of general tolerances, a rule that always decides was cheapest~~
1378 ~~where it was accurate, and an interval rule where a false accept is expensive relative~~
1379 ~~to a trial—, which differ by relation: a trial of a new force on an existing tool is a short~~
1380

press run, a trial of a new family may need a new tool. In a company, steps 1 to 5 fit the start of production or the production part approval of a family, when its series are measured anyway, and step 6 the later decisions on variants and process changes.

6.3 ~~Where simulation and machine learning help~~

6.3 Simulation and machine learning in this regime

The question the call for this collection poses about learning from manufacturing data receives a sober answer here. Within a design family, the production record carries the decision: the nearest produced setting is the most decisive rule, and adding the simulation to a calibrated rule does not improve it and, for the bias correction, makes it worse, because the simulation's process trends—the effect of blank-holder force above all—are not credible. A physically motivated calibration of carried the decision. The simulated force trend was 2.7 to 5.1 times too large, and calibrating the friction coefficient did not repair thisit, which points to the friction law and the blank-holder model rather than to a parameter. ~~Learning models, a Gaussian process or gradient boosting with or without the simulation, predict about as well as the nearest produced setting; what they add is an interval, which makes a rule~~ Learned models added an interval that is safe where the calibration alternatives are exchangeable with the new design and over-confident where they are not. The simulation is indispensable only for a new family, where nothing in the production record of another family carries over, but even there it supports decisions only at margins that a design team would not need a model for. A model's interval is a statement about the model; the safe distance for the evidence relation is a statement about production. A multi-task model cannot help a family with no produced alternative, because the correlation between families cannot be estimated without its data; the pooled Gaussian process, which carries the geometry as an input, is the borrowing of strength the data allow, and it improved coverage but not decisiveness. These findings concern one simulator of low fidelity for this production, used through offsets and one scale, and learners with six to nine produced settings each; a surrogate of these simulations inherits the simulator's distance, whatever its accuracy against the simulation. For the evaluation of AI-based DFM support ~~this means that accuracy on the lesson is methodological: held-out data accuracy is not enough;~~ a learned model should be compared with a production-record baseline ~~and~~, scored by evidence relation, in a unit that production defines, and shown to its user with a guard band. Whether that makes designers' reliance match a rule's reliability (Lee and See, 2004), and when an abstention should go to an expert rather than to a trial (Madras et al, 2018), are hypotheses for a study with design teams.

6.4 Implications for design research

The framework gives ~~three several~~ constructs of design research a production-referenced measure. ~~The safe distance of a rule for an evidence relation is the margin a design team has to keep between a prediction and a requirement for the verdict to be trusted, and so puts a number on what design margins are for (Eckert et al, 2019; Brahma and Wynn, 2020). The two distances decide~~ For variant,

adaptive and platform design (Pahl et al, 2007; Simpson et al, 2006), the distances
 by evidence relation quantify how far production knowledge carries from produced
 variants to a new one, with the nearest produced setting as the baseline that any
 capability database (Thornton, 2004) or learned model has to beat. The safe distance
 is the part of a margin that covers predictive uncertainty, as distinct from the
 part that absorbs change (Eckert and Isaksson, 2017; Eckert et al, 2019). For testing
 strategies (Thomke and Bell, 2001; Salado and Kannan, 2018; Camburn et al, 2017)
 overlapped development (Krishnan et al, 1997; Terwiesch et al, 2002)
 and set-based design (Sobek et al, 1999), the distances say when a test
 is cheaper than a decision, ~~the trade at the core of testing strategies~~
~~(Thomke and Bell, 2001; Salado and Kannan, 2018)~~ when design-stage information is
 fit to act on and how wide a feasibility limit has to be. For validation, the evidence
 relation is the question of the application domain (Oberkamp and Roy, 2010), and the
 calibration budget turns the production of variants into a planned verification activity.
 And they show how far simulation front-loads a decision (Thomke and Fujimoto, 2000)
 : only as far as its distance for the relation at hand distances could serve as evidence
 of a risk-informed credibility assessment (ASME, 2018). In the terms of the design
 research methodology (Blessing and Chakrabarti, 2009), the evaluation ~~reported here~~
 establishes that the framework discriminates between rules and evidence situations
 on real production data—its empirical performance is a support evaluation: it shows
 structural validity in the sense of the validation square (Seepersad et al, 2006)—while
 its use by design teams, and the success criteria that would link the distances to
 fewer trials or late changes, remain to be evaluated, that the metrics discriminate
 between rules and relations on real data (Seepersad et al, 2006), and is a first step
 towards performance validity, which would require showing that decisions taken with
 the framework are better than decisions taken without it.

6.5 ~~Applying the framework elsewhere~~Reuse for other processes and AI-based DFM tools

The framework needs ~~four things~~: several produced variants of a design family ,~~each~~
~~with whose relations to each other span those of the coming designs~~, a series of consec-
 utive parts or replicated batches of one design ~~from which the floor can be measured;~~
~~variants whose relations to each other span the relations of the coming designs; per~~
~~variant~~, a design-stage predictor for every variant, ~~which may be a simulation, and~~
~~characteristics measured identically on parts and predictions~~. Here the long series
 served the floor, whereas calibration runs of about 50 parts per variant qualified the
 rules as well as full series; how many long runs a floor needs was not tested. Table 6
 maps these requirements onto the processes named most often in DFM; attribute
 outcomes such as splits, porosity or short shots need a requirement on a rate rather
 than a quantile, which the framework does not yet cover. A predictor need not return
 an interval: any tool that returns meets, fails or trial for a stated requirement, such
 as a classifier at its operating threshold or a ~~surrogate or the production record~~
~~itself; and characteristics that can be measured identically on parts and predictions.~~
~~None is specific to forming~~ language-model check given the requirement value, can be
 scored by sweeping the requirement, and for continuous design spaces the relation

Table 6 What the framework requires, mapped onto other processes

Process	Batch and run	Dominant variation	Evidence relations	Floor
Sheet forming (this paper)	consecutive parts of one series	drift within a run; coil, die state	variant, setting, family	batch centres within a run
Injection moulding	shots of one cavity and set-up	cavity, shot, set-up	cavity, process window, new mould	within cavity, between cavities
Machining	consecutive parts between offsets	tool wear, compensated by offsets	tool path, cutting data, new part	between offset intervals
Casting	castings of one pattern and melt	melt, pattern position	alloy, gating, new pattern	between melts
Additive manufacturing	replicated specimens of one build	build, position, orientation	orientation, parameters, new part	between builds, via anchors

becomes a distance from the produced designs in descriptor space or the position relative to their hull. A minimal report comprises the floor with its protocol and variance components, the decisive and safe distances by relation with refitting intervals and one-sided values, the coverage of interval rules, the nearest-produced-setting and no-simulation baselines, and the guard bands and guard performance. Distances per characteristic rest on as many cases as there are produced alternatives, so a team would pool characteristics with similar ratios of floor to part scatter, or accumulate evaluations over families. An attempt to apply ~~it~~ the framework to an open dataset of powder bed fusion of PA12 (Leirimo, 2021) ~~showed what the first condition demands: grouping the failed on the second requirement: grouping~~ specimens by orientation class ~~would have mixed several mixed~~ orientations in a batch, and ~~only~~ eight identical anchor specimens per build ~~replicate one design. Two-sided requirements follow from the absolute deviation from the target, as for the arm angle and the dome; requirements on a high conformance share are scored in the same way as long as the series estimates the quantile, and need a parametric tail beyond it were too few for a floor; a between-build floor from replicated anchors would be the remedy.~~

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, and the relations within a family come from a 3×3 factorial in which ~~one factor, lubrication~~ ~~lubrication~~ is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two ~~calibrations~~ ~~transfers between geometries that differ in three features at once, and the intervals of its distances hold the two fits fixed.~~ Each alternative was produced once, so the floor covers drift within a run but not between days, coils or tool states; ~~the quasi-replicate series bound that variation from above, the reference q_{95} is that of one run, and the order and dates of the series are unknown; between-series variation bounds the missing component at 1.2–9.8 floors. No gauge study or acceptance test of the scanner (VDI/VDE, 2012; ?) was available: the redundancy of the scans bounds the independent measurement noise, but a drift of the scanner (Section 5.1). Scanner drift cannot be separated from that of the process, and systematic scan errors and differences of reference surface enter the absolute offsets of~~

the simulation, though not the floors and effects. The simulated characteristics carry extraction noise of up to several floors, and the documentation of the simulation leaves the material model open. The blank-holder force of 100 kN ran at a slower stroke. Lubrication is a pattern whose oil amount varies between series process drift. The extrapolation in force is confounded with stroke speed, which the simulation does not represent, so the extrapolation results describe a force and speed change together. The requirements are one-sided limits on the 95th percentile of a characteristic averaged over sides, the scenario covers two angles, and the reference q_{95} has a standard error of about 0.1 floor below the conformance of series release. Several rules, the grouped schemes and the estimator were added after the first guard bands were added or chosen after earlier results had been seen and are evaluated on the same data, so the comparison between rules carries selection optimism (Cawley and Talbot, 2010); the analysis was carried out by a single analyst, with code and data open for replication.

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1533 7 Conclusions

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Design-stage manufacturability decisions can be evaluated against production ~~in a unit on a scale~~ that production defines. ~~The production floor—the difference that drift and scatter produce between two batches of one design—expresses the fitness of any predictor for a decision independently of the predictor and the characteristic, and:~~ the decisive and safe distances ~~read as operating characteristics, in floors~~ tell how close to a requirement a rule decides and how close it can be ~~trusted relied on~~ not to mislead, ~~and a guard band turns them into a rule for a single decision.~~

On 18 deep-drawn design-process alternatives with matched simulations, the relation of a new design to the produced evidence governed decision fitness more than the model did. With best attainable distance grew with the novelty of the design: 3.4 floors with a produced sibling the production record decided within a few floors, 8.5 for a new process setting no rule decided closer than 8.5 floors, and, and no rule safe closer than 16 floors for a new family no rule was safe closer than 16 floors. The simulation contributed only to the new family; learned models matched the nearest produced setting in accuracy and added calibrated abstention where the calibration alternatives resembled the new design. Within a family the choice of rule mattered more than the relation, release-level requirements lay inside every decisive distance, and in this small-data regime the simulation as used, a model with a large and partly sign-inconsistent bias, contributed only across families, while learned models added abstention rather than accuracy.

Four extensions follow: series produced on several days, coils and tool states, so that the floor covers variation between runs; design families with continuous geometric descriptors, so that new designs can be placed between produced ones; a cost model that turns the distances into decisions on trials and margins; and an; a confirmatory evaluation of the procedure frozen procedure on another dataset or process; and a study with design teams of whether decisions taken with the framework are better. The released code, which reproduces every result from the public data, is the starting point for all four.

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Appendix A Notation, robustness and computation details

Table A1 summarises the notation of Section 3, and Table A2 gives the distances under the robustness variants; the other results quoted in the text, such as guard bands, coverage, paired differences and distances per characteristic, are written by the released code and archived with it.

Table A1 Notation and terms

Symbol, term	Meaning
$a, c, q_p(a, c)$	alternative (design–process combination), characteristic, p -quantile of the parts of a ($p = 0.95$)
R, d	requirement (upper limit on c or $ c $); its distance from the truth in floors, $R = q_p + dF_c$
$[L_r, U_r], V_r$	interval of rule r for q_p ; verdict meets, fails or trial (Eq. 1)
B, m, F_c	batch size (50); batch centre (10 % trimmed mean); production floor (Eq. 2)
σ_b, σ_w	standard deviation between batch means and between parts within a batch
ESR, MRC	effect-to-scatter ratio (Eq. 3); minimum resolvable change, $F_c/ s_c $
u, v	distances in floors by which an interval excludes the truth from above and below
$\kappa(\delta), \omega(\delta)$	shares of correct and wrong verdicts among the feasible verdicts at $ d = \delta$
δ_d, δ_s	decisive distance ($\kappa \geq 0.95$ beyond it) and safe distance ($\omega \leq 0.05$ beyond it)
FA, FR side	one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects)
g	guard band: observable margin beyond which at least 95 % of the decided verdicts were correct
Calibration alternatives	produced alternatives whose record calibrates a rule
Evidence relation	sibling (same setting produced), new process setting (interpolated, extrapolated), new family
Calibration budget	distances against the number k of produced alternatives, by relation
Guard	family, range and novelty checks that send a design outside the calibration to trial

Computation details. Batches need at least 80 % valid values. M5 and M5n are Gaussian-process regressions (constant times anisotropic squared-exponential kernel plus white noise, normalised targets, three restarts, length scales in $[0.1, 100]$) on a geometry indicator, the force in 100 kN and the lubrication rank. NN takes the mean q_{95} at the nearest force of the family, both neighbours when equally near. Mc calibrates, per lubrication pattern, the friction coefficient whose simulated draw-ins are closest to the calibration centres. M3 and M3n use histogram gradient boosting (depth 3, 200 iterations, rate 0.08) on geometry, force, lubrication, sheet thickness and oil film, with 500 incoming conditions drawn from calibration parts of the same pattern; their jackknife+ interval combines the leave-one-out predictions for the new alternative with the absolute leave-one-out residuals about their median. Guard bands lie on a grid of 0.25 floors. The refitting bootstrap draws the three lubrication patterns of each geometry with replacement 200 times, at least two of them distinct. Software: Python 3.11 with NumPy, pandas, SciPy, scikit-learn (Pedregosa et al, 2011) and Matplotlib.

Table A2 [Decisive/safe distances \(floors\) under the robustness variants](#)

Variant	New variant				New setting			New family	
	M1s	M2	M5	NN	M2	M5	NN	M2	M5
Reference	4.9	23/0.15	7.2/0.75	3.4	18/8.1	19/6.8	8.5	51/16	38/28
Temperature-adjusted	5.0	23/0.10	7.2/0.65	3.4	17/7.6	19/6.8	9.4	48/15	37/27
First 150 parts dropped	6.5	27/0.20	9.0/0.45	4.1	21/8.7	24/8.7	12	63/18	39/29
Floor between series	2.9	11/0	1.3/0.20	0.75	9.9/2.6	11/2.3	2.7	16/3.8	12/9.3
Floor: batches of 25	4.6	21/0.15	6.4/0.70	2.9	16/6.9	17/6.3	7.7	43/14	31/23
Floor: batches of 100	5.3	24/0.15	8.2/0.80	3.5	19/8.5	21/7.9	8.9	57/17	46/29
Floor: 90 % quantile	6.2	31/0.20	8.8/0.90	4.0	21/10	23/8.6	10	61/20	45/34
Floor: 99 % quantile	3.4	14/0.10	5.2/0.50	2.4	11/4.4	13/4.4	5.7	37/11	29/15
Calibration series of 50	5.0	22/0.20	6.8/0.70	3.4	18/7.5	19/6.2	8.4	50/16	37/28
Calibration series of 100	4.9	22/0.40	6.8/0.40	3.3	18/7.5	19/6.2	8.7	50/16	38/28
Calibration series of 250	4.9	22/0.45	7.0/0.45	3.3	18/7.5	19/6.2	8.7	50/16	38/28
Conformance $p = 0.90$	4.9	23/0.20	7.2/0.70	3.3	18/7.5	18/6.4	8.4	51/15	38/28
Conformance $p = 0.99$	5.1	23/0	7.5/0.70	3.3	18/8.3	20/6.9	8.7	51/16	38/28
Interval level 0.80		23/0.15	6.4/1.0		18/8.1	17/8.1		50/17	38/28
Interval level 0.95		23/0.15	8.0/0.25		18/8.1	20/6.2		51/16	39/27
Response surface	4.8	25/0.30	7.2/0.70	3.4	18/7.6	20/6.9	8.5	49/25	37/27
GP: sampling noise floor			6.7/1.4			18/7.4			38/30
GP: no noise floor			6.7/1.5			18/7.4			38/30
Unit: q_{95} reproducibility	5.3	17/0.15	6.9/0.40	2.8	15/6.0	18/6.2	6.9	41/13	34/30
Refit bootstrap, δ_d	4.9–7.3	22–23	4.0–11	2.3–4.6	16–18	18–20	6.9–9.2	49–53	34–39

[Rules refitted for every variant; empty cells: the variant does not affect the rule. Floor variants and the unit of \$q_{95}\$ reproducibility \(95 % quantile of the difference between the \$q_{95}\$ of two batches of 100 parts\) rescale the distances. Calibration series of \$n\$: the calibration alternatives' \$q_{95}\$ from their first \$n\$ parts. Conformance \$p\$: the decided statistic is the \$p\$ -quantile. Response surface: nominal simulation and envelope from a quadratic fit over thickness and friction. Other kernels and descriptors of the GPs move M5 by at most 0.4 floors. Refit bootstrap: 95 % interval when the lubrication patterns of each geometry are resampled and every rule is refitted \(200 resamples\).](#)

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Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. The extracted per-part and per-simulation feature tables are provided with the code.

Code availability. The complete analysis pipeline is released under an open licence at <https://github.com/htcanseven/Claude> (directory `scripts/`; ~~an archived release with a DOI will be deposited on Zenodo at acceptance~~) and reproduces every number in this paper from the public data; [a versioned archive of the code and the feature tables is deposited on Zenodo \[DOI pending\]](#).

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction ~~as an assistant~~ for code development, literature retrieval, data analysis ~~and~~, drafting and editing ~~of the text, and simulated peer review of earlier versions~~. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. ~~No AI-generated images are used; all~~ All figures are produced by the released code from the data.

2 Computation details and additional tables

1.1 Computation details

Feature extraction. Scans are reduced to a grid of 2×4 pixel medians after a luminescence filter (connected components of at least 20,040 pixels); calibration 0.0769 mm per pixel across the scan line, 0.1581 mm per line and 0.00775 mm per height unit. The cup-bottom plane is fitted robustly to the plateau at least 10 mm inside its edge, and all depths are taken from it. The flange comprises the points 20–40 mm below the bottom plane, at least 8 mm from the cup wall and 3 mm from the outer edge. Wall positions are interpolated where the binned (0.5 mm) radial profile reaches 10, 15 and 20 mm depth; the wall angle is taken between 10 and 20 mm. The arm angle is fitted to the longest run of the profile whose slope stays below 15° , shortened by 1 mm at both ends and at least 4 mm long. Flange widths are medians over bands ± 2 mm about the mid-lines; corner tips are the outline points within 0.3 mm of the extreme diagonal projection. Simulated parts are mirrored from the quarter model and processed with the same definitions on a 1 mm grid.

Floors and effects. Batches of $B = 50$ consecutive parts; batches with fewer than 80 % valid values are dropped; centres are 10 % trimmed means; $\alpha = 0.95$. Intervals: 2000 resamples of the batches within alternatives (seed 1), pairs of copies of one batch excluded, and 2000 resamples of whole alternatives. Scatter floor: 50 random permutations of the parts within each series. Components: rational subgroups of five consecutive parts; successive difference standard deviation $\sqrt{\Delta^2/2}$.

Decision rules. Adequacy share $p = 0.95$; per-geometry floors; requirements below zero are left out for these non-negative characteristics. The largest-residual margin is the $\lceil (n+1) \cdot 0.9 \rceil$ -th smallest absolute residual of n calibration alternatives, the largest when this exceeds n . M5 and M5n: Gaussian-process regression with a constant-times-anisotropic-squared-exponential kernel plus white noise, normalised targets, three optimiser restarts (random state 0), length scales in $[0.1, 100]$; the noise level is bounded below by the mean squared block-bootstrap standard error of the calibration alternatives' q_{95} relative to the variance of the targets; descriptors are a convex-geometry indicator, the blank-holder force in units of 100 kN and the lubrication rank (coarse 0, medium 1, fine 2, the order of the oil amount), with descriptors that are constant over the calibration set dropped. NN: mean q_{95} of the calibration alternatives of the target's geometry at the nearest blank-holder force (both

neighbours when equally near; the other geometry when the family has none). Mc: for each lubrication pattern, the friction coefficient of the DDACS grid at 0.99 mm whose simulated mid-side and corner draw-in, in floors, are closest in least squares to the centres of the calibration alternatives with that pattern (all calibration alternatives if none has it). M3 and M4: histogram gradient boosting (maximum depth 3, 200 iterations, learning rate 0.08, random state 0) on the geometry indicator, blank-holder force, two lubrication indicators, sheet thickness and oil film; 500 incoming conditions are drawn for the new alternative from calibration parts with the same lubrication pattern, with in-sample residuals pooled over the calibration alternatives resampled; offset and margin from a leave-one-out over the calibration alternatives.

Distances and intervals. The shares of correct and wrong verdicts are evaluated exactly on a grid of 0.05 floors up to 600 floors. Intervals: 1000 resamples of the held-out alternatives within each geometry (seeds derived from the scope), each with one of 1000 block-bootstrap replicates of every alternative's q_{95} (batches of 50 resampled within the series); the resamples are shared by all rules of a scope. Refitting bootstrap: 200 resamples of the alternatives within each geometry, every alternative-level rule refitted.

Budget, guard, scenarios, robustness. The budget uses all subsets of $k = 1, \dots, 8$ calibration alternatives of the family (M5 and M5n from $k = 2$), bands from 200 resamples of the held-out alternatives. The guards and the tolerance scenarios use the stored intervals of every rule. ISO 2768-1 angular rows by the shorter leg; the wall (about 28 mm) and the concave arm (about 18 mm) over 10 up to 50 mm; the convex arm (about 9 mm) up to 10 mm. The temperature correction subtracts $b(T - \bar{T}_a)$ from each characteristic, with b the pooled within-alternative slope on the punch temperature T . The smoothed simulation replaces the simulated values of each geometry and force by a quadratic response surface in sheet thickness and friction coefficient.

Software. Python 3.11 with NumPy 2.4, pandas 3.0, SciPy 1.17, scikit-learn 1.9 (Pedregosa et al, 2011) and Matplotlib 3.11.

1.1 Additional tables

Table ?? gives the bootstrap intervals of the decisive and safe distances, Table ?? the one-sided safe distances, Table ?? the coverage and width of the interval rules, Table ?? the paired differences between rules, Table ?? the distances per characteristic, Table ?? the part-level rules under other learners, Table ?? the hyperparameters of the Gaussian-process fits, Table ?? the redundancy of the scans and Table ?? the extraction noise of the simulations. Two analyses are provided with the code but not used here: the sensitivities of the simulation over the geometric corners of the simulation dataset, which have no production counterpart, and the in-line verifiability of the characteristics from the press-force record [no AI-generated images are used](#).

Bootstrap 95% intervals of the decisive and safe distances (floors)

⁰1000 resamples of the held-out alternatives within each geometry, each with a block-bootstrap replicate of the true q_{95} ; the fits are held fixed (refitting: Table A2). A new family rests on two calibrations, one per direction.

One-sided safe distances (floors): false accepts and false rejects	1749
Interval rules: coverage of the true q_{95} , interval width and the accuracy of the centre	1750
Paired differences of the decisive distance between rules (floors)	1751
Decisive/safe distance per characteristic (floors)	1752
Part-level rules M3 and M4 under other learners and settings (decisive/safe, floors)	1753
Gaussian-process fits with a hyperparameter at a bound	1754
Redundancy of the scans: ranges over the nine series of each geometry	1755
Extraction noise of the simulated characteristics (floors)	1756

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⁰ FA side: smallest distance beyond which at most 5 % of the verdicts on failing alternatives ($d < 0$) are false accepts; FR side: the same for false rejects of adequate alternatives ($d \geq 0$). Requirements below zero are left out.	1765
⁰ cov.: share of cases whose true q_{95} lies in the interval (%; nominal level 90 %; the maximum residual margin of M2 and M2n attains at most $n/(n+1)$ with n calibration alternatives); half: median half-width (floors); centre: decisive distance of the interval centre used as a point rule (floors).	1766
⁰ Difference and 95 % interval from the common resamples of Table ??; a negative difference means that the first rule decides closer to the requirement. M5–M5n, M2–M2n and M1–M1n isolate what the simulation adds to a rule.	1767
⁰ 18 held-out alternatives per characteristic and scope (36 verdicts per distance and side); one wrong verdict moves a share by 2.8 percentage points.	1768
⁰ Reference: histogram gradient boosting, depth 3, 200 iterations, rate 0.08. Regularised: depth 2, 100 iterations, rate 0.05, L2-penalty 1. Flexible: depth 6, 400 iterations, rate 0.1. Ridge: linear regression on the standardised inputs. Quantile loss: boosting of the 95th percentile, no resampled residuals. Selected: the boosting setting with the smallest leave-one-out error over the calibration alternatives.	1769
⁰ Share of the fits (calibration set and characteristic) whose noise level sits at its lower bound, the sampling variance of the calibration alternatives' q_{95} , and with at least one length scale at a bound (0.1 or 100 in units of 100 kN or one lubrication step).	1770
⁰ W_x, W_y : flange width across the cup centre, across and along the scan lines; E, W, N: east, west and north walls; SD: standard deviation over the parts of a series; correlation over the parts of a series.	1771
⁰ Noise: standard deviation of the second difference of the simulated value along the friction grid over $\sqrt{1.5}$ (white noise of the extraction). Envelope: largest difference between the maximum of the simulated values of a geometry and force and the maximum of a quadratic response surface over thickness and friction.	1772

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